

QUANTUM MACHINE LEARNING BASED MEDICAL IMAGE SEGMENTATION

Project Submitted By

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April 2026

Declaration

We hereby declare that this dissertation is the product of our own work, and we attest that it contains no material that resulted from collaboration, except where explicitly acknowledged in the text. Furthermore, we confirm that this project has not been previously submitted, either in part or in its entirety, to any other University or Institution for the purpose of obtaining any degree, diploma, or other qualification. All sources used and referenced in this dissertation are duly credited, and any borrowed ideas or information are appropriately cited in accordance with academic standards and guidelines.

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Certificate

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Abstract

In this thesis, we propose two novel quantum-classical hybrid architectures designed to address the challenges of tumor classification and tumor segmentation in MRI imaging. The first contribution introduces the *Quantum-Driven Probabilistic Feature Extraction (QDPFE)* framework, which overcomes the computational limitations and CNOT-heavy circuit depth inherent in NEQR-based quantum image encoding. Instead of encoding an entire image into a deep quantum circuit, QDPFE employs a scalable strategy by downsampling the image and dividing it into non-overlapping patches. Each patch is then transformed into a compact quantum representation using shallow, NISQ-friendly quantum circuits. A probabilistic thresholding is applied to measurement outcomes to extract discriminative and task-optimized feature vectors. These vectors are aggregated to form a global feature set that is subsequently fed into classical machine-learning classifiers. This hybrid approach significantly reduces qubit requirements, avoids decoherence-prone deep circuits, and provides a more practical alternative to NEQR for real-world medical imaging applications. As a *proof-of-concept*, the proposed QDPFE framework achieves an encouraging tumor classification accuracy of approximately 80% on a small MRI dataset, demonstrating the potential of probabilistic quantum feature extraction even under NISQ hardware constraints. The second contribution is a *novel Quantum U-Net* architecture developed for tumor segmentation and subtype classification. The method begins with training a multitasking U-Net capable of jointly learning segmentation masks and tumor-type labels using augmented MRI data. After the classical backbone is trained, the model is fine-tuned by integrating a quantum bottleneck module into the latent space of the network, enabling quantum-enhanced feature transformation and improved representation of complex tumor structures. Experimental results show that the Quantum U-Net achieves a Dice score of 79.16%, an Intersection over Union (IoU) of 70.26%, a precision of 80.96%, and a recall of 82.69% on the segmentation task, while attaining a high tumor-type classification accuracy of 97.39%. Notably, these improvements are achieved without a substantial increase in training time, indicating that quantum refinement enhances segmentation fidelity and classification robustness with minimal computational overhead. Overall, the proposed QDPFE and Quantum U-Net frameworks present a practical, scalable, and NISQ-compatible pathway for integrating quantum computation into medical image analysis, offering compelling evidence that carefully designed hybrid architectures can deliver meaningful performance gains in tumor classification and segmentation.

Chapter 1

Introduction

Medical imaging has become a cornerstone of modern healthcare, enabling non-invasive visualization of anatomical structures and pathological conditions with remarkable precision. Among the various imaging modalities, Magnetic Resonance Imaging (MRI) is particularly significant in neuro-oncology for its superior soft tissue contrast and detailed visualization of brain structures [1]. MRI plays a vital role in the detection, localization, and characterization of brain tumors, providing clinicians with essential information for diagnosis, treatment planning, and prognosis assessment [2]. Brain tumors, characterized by uncontrolled and abnormal cellular proliferation, may arise from the brain parenchyma itself (primary tumors) or metastasize from other organs (secondary tumors). Depending on their aggressiveness and growth rate, tumors are classified into benign and malignant categories, where malignant tumors pose severe clinical challenges due to their infiltrative nature and poor survival outcomes. The accurate classification and segmentation of brain tumors from medical images are crucial steps in computer-aided diagnosis (CAD) systems. Segmentation refers to the process of delineating regions of interest (ROI) within medical images, such as isolating tumor boundaries from surrounding healthy tissue. Precise segmentation serves as the foundation for further analysis, including feature extraction and classification. However, achieving robust segmentation remains a major challenge due to factors such as image noise, intensity inhomogeneity, and irregular tumor morphology. Variations in scanner types, acquisition parameters, and patient-specific anatomy further complicate automated segmentation efforts. These complexities demand sophisticated algorithms capable of adapting to heterogeneous data while maintaining high spatial accuracy and consistency.

Once segmentation is achieved, extracted features—typically encompassing texture, intensity, and shape descriptors—play a pivotal role in differentiating between normal and pathological tissues. Texture-based descriptors such as the Gray Level Co-occurrence Matrix (GLCM) quantify spatial variations in pixel intensity, which are particularly informative for distinguishing between benign and malignant lesions. In one study, the use of 512-dimensional GLCM features combined with Random Forest classification achieved 84% accuracy when using the entire tumor volume and 74% accuracy when features were restricted to the central tumor region, validated using tenfold cross-validation [3]. These findings underscore the discriminative potential of texture and spatial correlation features in tumor characterization.

Tumor classification, the subsequent stage following segmentation and feature extraction, aims to categorize tumor types—typically into benign or malignant—based on the derived quantitative features. Machine learning (ML) and, more recently, deep learning (DL) models have revolutionized this process by learning complex nonlinear mappings between imaging features and pathological outcomes. Classical ML models such as Support Vector Machines (SVM) [4], Random Forests [5], and k-Nearest Neighbors (kNN) [6] rely on handcrafted features, while modern deep learning architectures such as Convolutional Neural Networks (CNNs) [7] automatically learn hierarchical representations directly from image data. These models have demonstrated state-of-the-art performance in medical image analysis, reducing dependency on manual intervention and improving reproducibility across datasets.

Segmentation and classification processes are deeply interdependent: precise segmentation enhances feature reliability, thereby improving classification accuracy. Conversely, effective classification algorithms can guide segmentation by refining the delineation of pathological regions. Despite remarkable progress, challenges persist—particularly in addressing the variability of tumor appearance across patients, dealing with limited annotated datasets, and mitigating computational complexity. The ongoing research in multimodal imaging integration, where information from modalities such as MRI, CT, and PET is fused, further enhances tumor characterization by leveraging complementary spatial and functional information [8].

1.1 Machine Learning

Machine learning (ML) has emerged as a transformative technology in medical image analysis, offering computational frameworks capable of automatically identifying complex patterns in imaging data that may not be discernible through manual interpretation. In the context of tumor classification and segmentation, ML algorithms leverage statistical learning and optimization principles to establish mappings between image-derived features and diagnostic outcomes [9]. This approach enables the automatic discrimination between normal and abnormal tissues, as well as the delineation of tumor regions with high precision, thereby supporting clinical decision-making. The traditional ML workflow for medical image-based tumor analysis generally involves four primary stages: preprocessing, feature extraction, feature selection, and classification. Preprocessing addresses the inherent challenges in medical images, such as noise, low contrast, and non-uniform illumination, through methods like histogram equalization, Gaussian filtering, and bias field correction. [10] Once the data are preprocessed, quantitative features are extracted to represent the intrinsic properties of the segmented tumor region. These features often encompass intensity, texture, and shape characteristics, where texture descriptors such as the Gray Level Co-occurrence Matrix (GLCM) [3], Local Binary Patterns (LBP) [11], and wavelet coefficients [12] are widely utilized for their ability to capture spatial heterogeneity and microstructural variations in tumor tissues.

Feature selection plays a critical role in reducing dimensionality and improving the generalization capacity of ML models by identifying the most informative and non-redundant features. Statistical approaches like Recursive Feature Elimination (RFE), Principal Component Analysis (PCA), and mutual information-based selection are often employed to eliminate irrelevant variables, thereby optimizing model performance and computational efficiency [13].

Subsequent to feature selection, classification algorithms are applied to categorize tumor types, grades, or regions. Among the classical ML methods, Support Vector Machines (SVM) [4], Random Forests (RF) [5], Decision Trees (DT) [14], k-Nearest Neighbors (k-NN) [6], and Naïve Bayes classifiers [15] are extensively used in tumor detection tasks. These algorithms operate on different mathematical foundations—SVMs utilize hyperplane-based separation in high-dimensional space, Random Forests rely on ensemble learning through multiple decision trees, and k-NN uses proximity-based classification determined by feature space similarity. For instance, a Random Forest classifier trained on 512-dimensional GLCM features has demonstrated 84% accuracy in full-tumor segmentation and 74% accuracy for central tumor regions using tenfold cross-validation [3], highlighting the effectiveness of ensemble learning for high-dimensional texture analysis.

In tumor segmentation, ML algorithms are employed either directly or in combination with region-based or clustering techniques. Supervised learning approaches use annotated datasets where tumor boundaries are manually delineated, allowing algorithms to learn discriminative patterns between tumor and non-tumor tissues. Conversely, unsupervised techniques, such as k-means [16] and fuzzy c-means clustering [17], which can be particularly useful when labeled datasets are scarce. Semi-supervised and active learning methods further bridge this gap by incorporating limited expert annotations into large unlabeled datasets, improving segmentation performance in real-world clinical settings. The integration of ML into medical imaging has significantly advanced automated tumor classification and segmentation; however, its performance is highly dependent on data quality, representativeness, and the robustness of feature engineering. Current research trends are oriented toward hybrid approaches that combine classical ML techniques with deep learning architectures, exploiting the interpretability of handcrafted features and the adaptability of neural networks. Such hybrid models are increasingly effective in addressing the limitations of conventional methods, paving the way toward accurate, efficient, and clinically viable solutions for medical image-based tumor analysis.

1.2 Deep Learning

Deep learning (DL), a subfield of machine learning, has revolutionized medical image analysis by enabling automatic feature learning from raw imaging data. Unlike traditional machine learning models that depend on handcrafted features, deep learning architectures learn hierarchical representations directly from pixel-level information, thereby capturing intricate spatial and contextual relationships inherent in medical images. In tumor classification and segmentation

tasks, deep learning models have demonstrated remarkable success due to their capability to generalize complex patterns and adapt to diverse imaging modalities such as MRI, CT, and PET.

Among various architectures, Convolutional Neural Networks (CNNs) serve as the foundational framework for deep learning in medical imaging. CNNs employ convolutional and pooling layers to extract local and global features, enabling the model to identify tumor regions, edges, and textural variations effectively. However, conventional CNNs often face limitations in preserving fine-grained spatial details and handling irregular tumor boundaries, especially in medical images where tumor regions exhibit high variability in intensity, shape, and size [7]. To overcome these limitations, several advanced architectures such as U-Net [18], ResU-Net [19], Swin U-Ne [20], and YOLO-based networks [21] have been developed, each optimized for precise segmentation and detection tasks.

1.3 Quantum Computing

Quantum computing represents a revolutionary advancement in computational technology, leveraging the fundamental principles of quantum mechanics—notably superposition, entanglement, and interference—to perform operations far beyond the capabilities of classical systems [22]. Unlike classical computers, which process information using bits that exist in a binary state of either 0 or 1, quantum computers use qubits (quantum bits), which can exist simultaneously in multiple states. This phenomenon, known as superposition, enables quantum computers to perform a vast number of calculations in parallel, drastically increasing computational efficiency and speed.

Another core property, entanglement, allows qubits to be correlated with each other such that the state of one qubit directly influences the state of another, regardless of distance. This interconnectedness enhances information processing capacity and ensures that computations account for complex interdependencies within data. The third principle, quantum interference, enables quantum systems to amplify the probability of correct outcomes while minimizing errors through constructive and destructive interference. Together, these properties provide the foundation for a new form of computation capable of addressing challenges that are computationally infeasible for classical architectures.

In the context of medical image processing, quantum computing offers transformative potential. Biomedical images such as MRI, CT, and PET scans contain enormous volumes of high-dimensional data that are computationally intensive to analyze. Classical algorithms often struggle with processing speed, memory constraints, and noise sensitivity when dealing with such large and complex datasets. Quantum computing, however, can exploit quantum parallelism to process entire image datasets simultaneously rather than sequentially, leading to significant acceleration in computation time. This advantage is particularly critical in diagnostic imaging tasks, where rapid and accurate image reconstruction, enhancement, and analysis are essential.

Moreover, quantum algorithms can handle high-dimensional vector spaces efficiently, making them ideally suited for complex image transformations, filtering, and reconstruction operations. For instance, quantum Fourier transforms and quantum wavelet transforms can be applied to image frequency analysis and compression with exponentially reduced computational costs [23]. Quantum-based image representation also allows for compact storage of multidimensional image data using fewer qubits, which optimizes both memory usage and data transmission. Noise reduction and denoising are other areas where quantum computing shows promise. Quantum superposition enables simultaneous examination of multiple image states, allowing enhanced noise filtering while preserving fine structural details a challenge in conventional image processing. Additionally, quantum entanglement facilitates correlation-based analysis between image pixels or voxels, improving the accuracy of pattern recognition and anomaly detection in medical scans.

Recent advancements in quantum hardware—such as increased qubit fidelity, improved coherence times, and scalable architectures—have brought practical implementations of quantum algorithms closer to realization [24]. As quantum gate operations become more stable and error correction improves, quantum systems are expected to perform complex image operations, including segmentation, enhancement, and 3D reconstruction, with unprecedented precision.

In essence, quantum computing introduces a fundamentally new computational framework capable of revolutionizing medical image processing. By utilizing superposition for parallel computation, entanglement for correlated data analysis, and interference for error minimization, quantum computing has the potential to dramatically improve the speed, efficiency, and accuracy of medical diagnostics. This emerging technology thus represents a crucial step toward next-generation computational medicine, where large-scale biomedical.

1.4 Quantum Machine Learning

Quantum Machine Learning (QML) has emerged as a powerful interdisciplinary field that combines principles of quantum computation with the learning capabilities of classical machine learning. Unlike conventional computation, which operates on binary logic, quantum computation leverages quantum mechanical properties such as superposition, entanglement, and tunneling. These properties allow quantum models to explore high-dimensional spaces more efficiently and provide new ways of representing and processing complex data. As a result, QML has the potential to achieve faster computation, improved pattern recognition, and enhanced performance for tasks involving large and heterogeneous datasets.

In recent years, QML has attracted substantial attention across diverse scientific domains, including chemistry, physics, industry, and healthcare. In the biomedical field in particular, the explosion of data from imaging systems, omics technologies, biosignal acquisition, and electronic health records has created an urgent need for advanced computational tools capable

of handling high complexity and variability. QML offers a promising avenue to address these challenges by enabling more efficient feature extraction, classification, prediction, and decision-making processes.

Medical imaging is one of the most active areas for QML applications. Modern imaging modalities—such as MRI, CT, PET, and microscopy—produce high-resolution, multi-channel data that demand significant computational resources for analysis. QML techniques can help accelerate tasks such as image classification, segmentation, anomaly detection, and disease prediction by exploiting quantum-enhanced models. Approaches like quantum neural networks [25], quantum support vector machines [26], quantum convolutional networks [27], and hybrid quantum-classical architectures aim to capture subtle and complex patterns that may be difficult for classical algorithms to distinguish. This can support physicians and biomedical practitioners by reducing diagnostic errors, improving prognostic accuracy, and enabling faster processing of large imaging datasets.

Overall, QML represents a significant opportunity to advance biomedical data analysis, particularly in medical imaging, where high-dimensional and information-rich data continue to grow. As research progresses, QML is expected to play an increasingly important role in developing efficient, accurate, and scalable healthcare solutions.

1.5 Motivation

The motivation behind this research work is to overcome classical limitations in capturing complex MRI patterns while enabling efficient learning on limited medical datasets. By integrating quantum encoding with deep segmentation models, the research aims to enhance tumor classification accuracy, improve segmentation fidelity, and explore practical quantum advantages in medical imaging.

1.6 Contributions

This work introduces a novel QDPFE framework whose qubit requirement depends only on patch size, unlike NEQR whose qubits scale impractically with full-image resolution. QDPFE uses shallow, NISQ-friendly circuits, performs task-optimized feature extraction, and integrates smoothly with classical models, avoiding NEQR's deep, CNOT-heavy and hardware-limited encoding.

The proposed Quantum U-Net further contributes improved Dice scores and superior tumor-type classification accuracy while maintaining comparable training time. This demonstrates that quantum refinement enhances representational power without significant computational overhead, offering a novel, practical pathway for hybrid quantum–classical medical imaging.

1.7 Organization of the Chapters

The remainder of this paper is organized as follows. **Section Introduction** presents the introduction and motivation of the study. **Background Study** provides the necessary background, followed by **Section Related Work** which reviews related research in quantum and medical imaging domains. **Section Methodology** describes both proposed methodologies, QDPFE and the Quantum U-Net framework, in detail. **Results** reports the experimental results and **analysis**. Finally, **Section Conclusion & Future Work** concludes the paper and outlines future research directions.

Chapter 2

Background Study

2.1 Machine Learning Models

2.1.1 Support Vector Machines

Support Vector Machine (SVM) [4] is a supervised learning algorithm used for classification and regression tasks. It aims to find an optimal hyperplane that separates data points of different classes with the maximum margin.

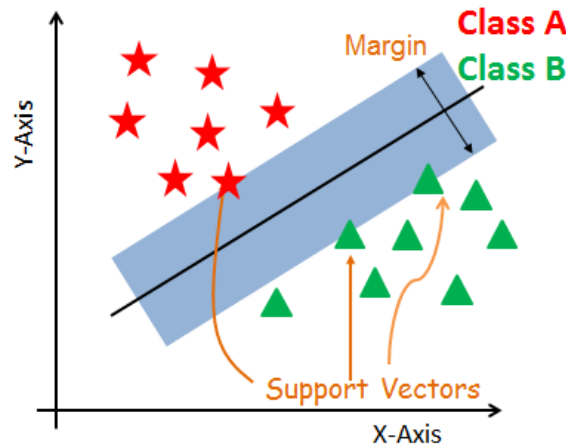


Figure 2.1: Decision Boundary in SVM

Mathematical Formulation

Given a training dataset (x_i, y_i) , where $x_i \in \mathbb{R}^n$ and $y_i \in \{-1, +1\}$, the SVM optimization problem can be formulated as:

$$\min_{w, b} \frac{1}{2} \|w\|^2 \quad (2.1)$$

subject to

$$y_i(w^T x_i + b) \geq 1, \quad \forall i \quad (2.2)$$

Here, w is the weight vector defining the hyperplane, and b is the bias term. The decision function is expressed as:

$$f(x) = \text{sign}(w^T x + b) \quad (2.3)$$

Kernel Function

When data are not linearly separable, SVM employs a *kernel function* $K(x_i, x_j)$ to map data into a higher-dimensional feature space without explicit transformation, a concept known as the *kernel trick*. Commonly used kernels include:

- **Linear:** $K(x_i, x_j) = x_i^T x_j$
- **Polynomial:** $K(x_i, x_j) = (x_i^T x_j + c)^d$
- **Radial Basis Function (RBF):** $K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$

This kernel approach enables SVMs to capture complex and nonlinear relationships, making them effective in domains such as medical imaging and biomedical data analysis.

2.1.2 Random Forest (RF)

Random Forest (RF) [28] is an ensemble learning algorithm used for classification and regression tasks. It builds multiple decision trees and combines their outputs to improve accuracy and reduce overfitting.

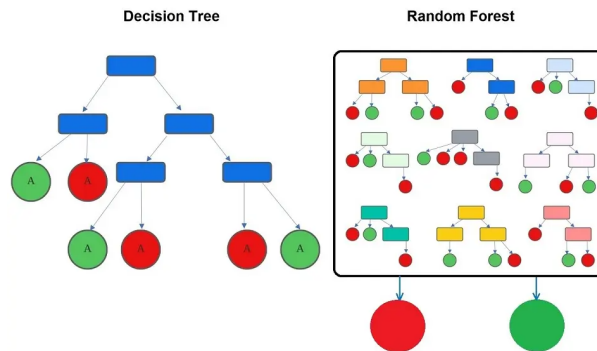


Figure 2.2: Random Forest

Mathematical Formulation

Given a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, a Random Forest consists of T trees $\{h_t(x)\}_{t=1}^T$. For classification:

$$\hat{y} = \text{mode}\{h_t(x)\}_{t=1}^T \quad (2.4)$$

For regression:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T h_t(x) \quad (2.5)$$

Working Principle

Each tree is trained on a random subset of data and features (bootstrap sampling). Final prediction is obtained by majority voting (classification) or averaging (regression). Random Forest provides robustness, handles nonlinearity, and performs well on biomedical and medical imaging tasks.

2.1.3 Logistic Regression (LR)

Logistic Regression (LR) [29] is a supervised learning algorithm commonly used for binary classification. It models the probability that a given input belongs to a particular class using a logistic (sigmoid) function.

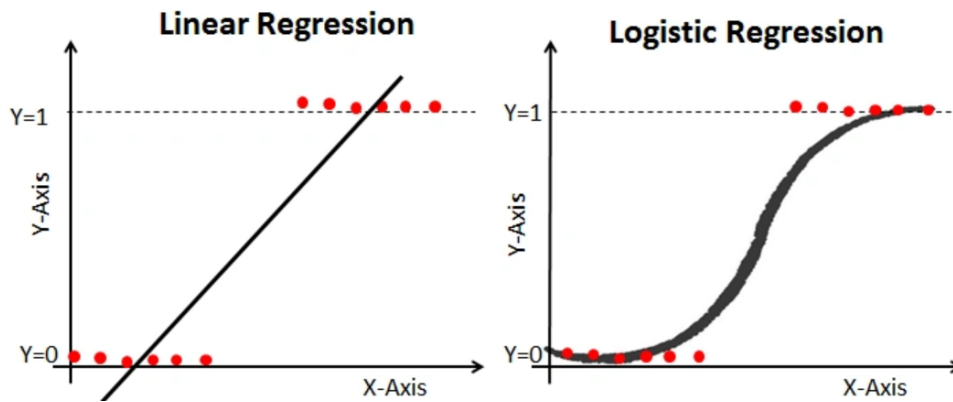


Figure 2.3: Linear v/s Logistic Regression

Mathematical Formulation

Given input features $x \in \mathbb{R}^n$, the model predicts the probability of class $y = 1$ as:

$$P(y = 1|x) = \sigma(w^T x + b) = \frac{1}{1 + e^{-(w^T x + b)}} \quad (2.6)$$

where w represents the weight vector and b is the bias term.

The decision rule is defined as:

$$\hat{y} = \begin{cases} 1, & \text{if } P(y = 1|x) \geq 0.5 \\ 0, & \text{otherwise} \end{cases} \quad (2.7)$$

Working Principle

Model parameters (w, b) are optimized by minimizing the cross-entropy loss:

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (2.8)$$

Logistic Regression is simple, interpretable, and performs effectively on linearly separable biomedical and medical imaging data.

2.1.4 K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) [6] is a simple, non-parametric supervised learning algorithm used for classification and regression. It makes predictions based on the majority class or average value of the k closest samples in the feature space.

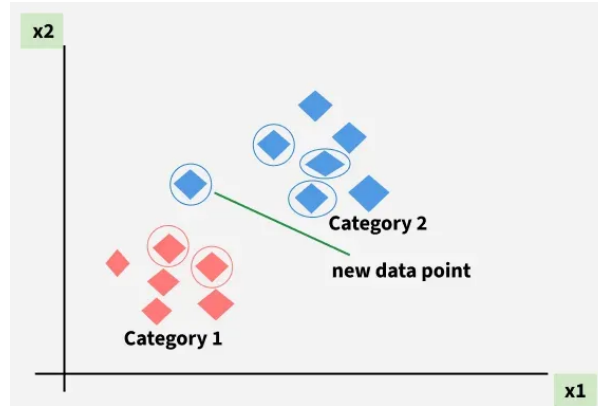


Figure 2.4: KNN

Mathematical Formulation

Given a training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ and a query sample x , the distance between x and each training point is computed, commonly using the Euclidean distance:

$$d(x, x_i) = \sqrt{\sum_{j=1}^n (x_j - x_{ij})^2} \quad (2.9)$$

The k samples with the smallest distances form the neighborhood $\mathcal{N}_k(x)$. For classification:

$$\hat{y} = \text{mode}\{y_i \mid x_i \in \mathcal{N}_k(x)\} \quad (2.10)$$

For regression:

$$\hat{y} = \frac{1}{k} \sum_{x_i \in \mathcal{N}_k(x)} y_i \quad (2.11)$$

Working Principle

KNN makes no assumptions about data distribution and relies purely on similarity measures. It performs well on small datasets and is often used in biomedical and medical imaging tasks for pattern recognition and disease classification.

2.1.5 Gradient Boosting (GB)

Gradient Boosting (GB) [30] is an ensemble learning algorithm that builds a strong predictive model by combining multiple weak learners, typically decision trees. Each new tree is trained to correct the errors made by the previous ensemble through gradient-based optimization.

Mathematical Formulation

Given training data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ and a loss function $L(y, F(x))$, Gradient Boosting builds the model iteratively as:

$$F_m(x) = F_{m-1}(x) + \eta h_m(x) \quad (2.12)$$

where $F_{m-1}(x)$ is the model from the previous iteration, $h_m(x)$ is the newly added weak learner, and η is the learning rate controlling the contribution of each tree.

At each stage, $h_m(x)$ is fitted to the negative gradient of the loss function:

$$r_{im} = - \left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)} \right]_{F(x)=F_{m-1}(x)} \quad (2.13)$$

Working Principle

The algorithm starts with an initial prediction and sequentially adds trees that minimize the residual errors. By combining many weak learners, Gradient Boosting achieves high accuracy and robustness, making it effective in biomedical data classification, disease prediction, and medical image analysis tasks.

2.2 Deep Learning Architecture

2.2.1 U-Net Architecture

The U-Net architecture, proposed by Ronneberger et al [18]. in 2015, is one of the most influential deep learning models for medical image segmentation. It adopts an encoder–decoder structure, where the encoder path captures semantic information through successive convolutions and down-sampling, and the decoder path reconstructs spatial resolution using up-sampling and skip connections. These skip connections bridge corresponding encoder and decoder layers, enabling the network to retain fine boundary details that are often lost during down-sampling. U-Net has been extensively utilized in brain tumor segmentation tasks due to its ability to accurately delineate tumor boundaries even with limited annotated data. The symmetric design and efficient use of contextual information make it particularly effective for pixel-level classification in MRI-based segmentation.

2.2.2 ResU-Net Architecture

ResU-Net [19] enhances the traditional U-Net by integrating residual learning, as introduced in the ResNet framework. In ResU-Net, residual blocks are incorporated within the encoder and decoder paths, allowing the model to learn identity mappings and mitigate the vanishing gradient problem common in deep networks. This design promotes better feature propagation and model convergence, especially for complex medical images with heterogeneous intensity distributions. The inclusion of residual connections also improves robustness against noise and facilitates deeper network training without performance degradation. In tumor segmentation, ResU-Net demonstrates superior boundary preservation and improved Dice similarity coefficients compared to conventional U-Net, making it a preferred choice for 3D MRI and CT segmentation tasks.

2.2.3 Swin U-Net Architecture

The Swin U-Net [20], a transformer-based extension of U-Net, introduces a novel attention mechanism derived from the Swin Transformer architecture. Instead of relying solely on convolutional operations, Swin U-Net employs hierarchical self-attention mechanisms that partition images into non-overlapping windows, enabling the model to capture both local and global dependencies efficiently. This hierarchical representation allows the network to model long-range contextual relationships, which is crucial for distinguishing tumor regions from surrounding healthy tissues that exhibit similar intensity patterns. Swin U-Net achieves outstanding performance in brain tumor segmentation benchmarks such as the BRATS dataset, outperforming traditional CNN-based models by effectively balancing spatial precision and contextual understanding.

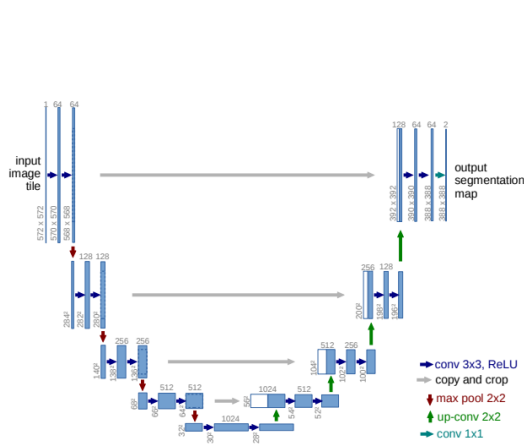


Figure 2.5: U-Net proposed by Ronneberger

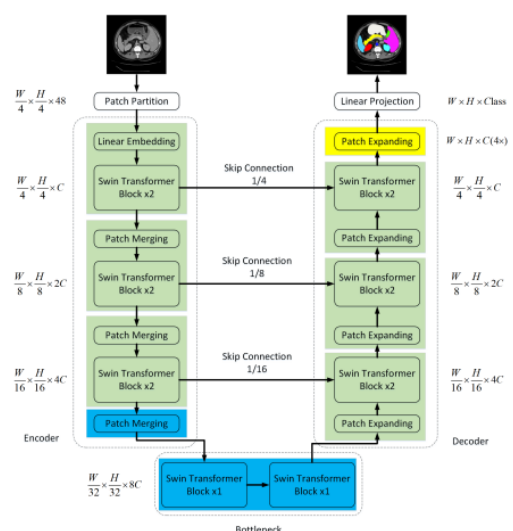


Figure 2.6: Swin U-Net proposed by Cao et al.

2.3 Basics of Quantum Computing

2.3.1 Qubit

Definition:

A *Quantum Bit (Qubit)* is a quantum mechanical system that can exist in an infinite number of states. A single qubit state can be specified by a pair of complex numbers (a, b) where $|a|^2 + |b|^2 = 1$.

Examples:

- The space of all possible polarization states of a photon is a qubit state.
- The space of all possible spin states of an electron is a qubit state.

Single Qubit State Space

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ b \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \end{pmatrix} = a|0\rangle + b|1\rangle$$

where

$$|a|^2 + |b|^2 = 1, \quad a = a_0 + ia_1, \quad b = b_0 + ib_1, \quad |x| = \sqrt{x_r^2 + x_i^2}, \text{ where } x = x_r + ix_i.$$

Vector Space

A single qubit state space is a two-dimensional vector space over the field of complex numbers $\mathbb{C}$. We represent it as:

$$\mathbb{C}^2 = \left\{ \begin{pmatrix} a \\ b \end{pmatrix} : a, b \in \mathbb{C} \text{ such that } |a|^2 + |b|^2 = 1 \right\}$$

Computational Basis:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Hence, a qubit state is written as:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Notes:

- A single qubit state can always be expressed as a linear combination of $|0\rangle$ and $|1\rangle$.
- It is possible to *set* a single qubit to a particular qubit state.

- However, *reading* (or cloning) a qubit state is not possible due to the no-cloning theorem, and storing a qubit indefinitely is currently technologically infeasible.

2.3.2 Superposition of States

A qubit can exist in a combination of two basis states simultaneously, which is known as a **superposition**. The general form of a qubit state is given by:

$$|\psi\rangle = a|0\rangle + b|1\rangle, \quad \text{where } |a|^2 + |b|^2 = 1$$

If both $a \neq 0$ and $b \neq 0$, the qubit is said to be in a **superposition** of the two states $|0\rangle$ and $|1\rangle$. The pair $\{|0\rangle, |1\rangle\}$ forms the **computational basis** of the single-qubit state space. This basis is also known as the *standard basis*, since every other single-qubit state can be expressed as a linear combination of these two.

Note (in red): *Once a superposition is not always a superposition.* That is, a state that is in superposition with respect to one basis may not be in superposition with respect to another.

For example, consider the quantum state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

This state is in superposition with respect to the computational basis $\{|0\rangle, |1\rangle\}$. However, with respect to the **Hadamard basis** $\{|+\rangle, |-\rangle\}$, it is simply $|+\rangle$, which means it is **not** in superposition in that basis.

2.3.3 Hadamard Basis

The Hadamard (or diagonal) basis is defined as:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Change of Basis Representation

Suppose a quantum state is written in the computational basis as:

$$|\psi_1\rangle = a|0\rangle + b|1\rangle$$

We can also express it in the Hadamard basis as:

$$|\psi_2\rangle = c|+\rangle + d|-\rangle$$

Expanding $|\psi_2\rangle$ in terms of the computational basis gives:

$$|\psi_2\rangle = \frac{c}{\sqrt{2}}(|0\rangle + |1\rangle) + \frac{d}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{c+d}{\sqrt{2}}|0\rangle + \frac{c-d}{\sqrt{2}}|1\rangle$$

By comparison with $|\psi_1\rangle = a|0\rangle + b|1\rangle$, we obtain:

$$\frac{c+d}{\sqrt{2}} = a, \quad \frac{c-d}{\sqrt{2}} = b$$

Solving for c and d gives:

$$c = \frac{a+b}{\sqrt{2}}, \quad d = \frac{a-b}{\sqrt{2}}$$

Thus, the same qubit state can be expressed as:

$$|\psi_1\rangle = \frac{a+b}{\sqrt{2}}|+\rangle + \frac{a-b}{\sqrt{2}}|-\rangle$$

This illustrates that a state which appears as a superposition in one basis may appear as a single basis vector in another — emphasizing that superposition is *basis-dependent*.

2.3.4 Complex Inner Product

Let the n -dimensional quantum state vectors be defined as:

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}, \quad |b\rangle = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

Then, the **complex inner product** (also called the *bra-ket* inner product) between these two vectors is defined as:

$$\langle a|b\rangle = \begin{pmatrix} \overline{a_1} & \overline{a_2} & \cdots & \overline{a_n} \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} = \sum_{i=1}^n \overline{a_i} b_i$$

If we take a single vector $|x\rangle = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$, then the inner product of the vector with itself is given

by:

$$\langle x|x \rangle = \overline{x_1}x_1 + \overline{x_2}x_2 + \cdots + \overline{x_n}x_n = |x_1|^2 + |x_2|^2 + \cdots + |x_n|^2$$

Thus, $\langle x|x \rangle$ represents the **squared magnitude (norm)** of the vector $|x\rangle$ in the complex vector space.

Measurement of a Single Qubit

Any measurement of a quantum system is associated with an **orthonormal basis** of its state space.

Let the quantum state of a single qubit be:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

where $a, b \in \mathbb{C}$ and $|a|^2 + |b|^2 = 1$.

When a measurement is performed in the computational basis $\{|0\rangle, |1\rangle\}$, the possible outcomes are the states $|0\rangle$ and $|1\rangle$, with probabilities given by:

$$P(0) = |\langle 0|\psi\rangle|^2 = |a|^2, \quad P(1) = |\langle 1|\psi\rangle|^2 = |b|^2$$

Hence, the qubit collapses to $|0\rangle$ or $|1\rangle$ with probabilities $|a|^2$ and $|b|^2$ respectively.

2.3.1 Orthonormality

A set of basis vectors $\{|\psi_1\rangle, |\psi_2\rangle\}$ is said to be **orthonormal** if:

$$\langle \psi_1|\psi_1 \rangle = 1, \quad \langle \psi_2|\psi_2 \rangle = 1, \quad \langle \psi_1|\psi_2 \rangle = 0, \quad \langle \psi_2|\psi_1 \rangle = 0$$

Example 1: Computational Basis

For the computational basis $\{|0\rangle, |1\rangle\}$:

$$\langle 0|0 \rangle = 1, \quad \langle 0|1 \rangle = 0, \quad \langle 1|0 \rangle = 0, \quad \langle 1|1 \rangle = 1$$

Thus, the computational basis is orthonormal.

Example 2: Hadamard Basis

The Hadamard basis vectors are defined as:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Now, verify orthonormality:

$$\langle +|+\rangle = \frac{1}{2}(\langle 0| + \langle 1|)(|0\rangle + |1\rangle) = \frac{1}{2}(\langle 0|0\rangle + \langle 0|1\rangle + \langle 1|0\rangle + \langle 1|1\rangle) = \frac{1}{2}(1 + 0 + 0 + 1) = 1$$

Similarly,

$$\langle -|-\rangle = 1$$

and

$$\langle +|-\rangle = \frac{1}{2}(\langle 0| + \langle 1|)(|0\rangle - |1\rangle) = \frac{1}{2}(\langle 0|0\rangle - \langle 0|1\rangle + \langle 1|0\rangle - \langle 1|1\rangle) = \frac{1}{2}(1 - 0 + 0 - 1) = 0$$

Hence, the Hadamard basis $\{|+\rangle, |-\rangle\}$ is also orthonormal.

2.3.2 Transformation of a Single Qubit State

A single qubit state can be expressed as:

$$|\psi\rangle = \begin{bmatrix} a \\ b \end{bmatrix} = a|0\rangle + b|1\rangle$$

Quantum gates perform linear transformations on such state vectors. For example, the transformation

$$\begin{bmatrix} a \\ b \end{bmatrix} \longrightarrow \begin{bmatrix} b \\ a \end{bmatrix}$$

is represented by the matrix:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

which is known as the **Pauli-X Gate**.

Matrix Representation of Outer Products

$$|0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad |0\rangle\langle 1| = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad |1\rangle\langle 0| = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Hence, the Pauli-X gate can also be written as:

$$X = |0\rangle\langle 1| + |1\rangle\langle 0|$$

Effect of X Gate on a General Qubit

Let

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Then,

$$X|\psi\rangle = (|0\rangle\langle 1| + |1\rangle\langle 0|)(a|0\rangle + b|1\rangle) = a|1\rangle + b|0\rangle$$

Thus, the X gate swaps the amplitudes of $|0\rangle$ and $|1\rangle$, analogous to the classical NOT operation:

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle$$

Unitary Transformations

Only **unitary transformations** can be implemented on a quantum computer. A transformation A is unitary if:

$$A^{-1} = A^\dagger$$

where $A^\dagger$ denotes the conjugate transpose of A .

Hadamard Transformation

The **Hadamard gate (H)** transforms basis states as:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle\langle 0| + |1\rangle\langle 0| + |0\rangle\langle 1| - |1\rangle\langle 1|)$$

$$H|0\rangle = |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Pauli Transformations

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = |0\rangle\langle 1| + |1\rangle\langle 0|$$

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = -i|1\rangle\langle 0| + i|0\rangle\langle 1|$$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = |0\rangle\langle 0| - |1\rangle\langle 1|$$

These three gates (X, Y, Z) form the fundamental set of **Pauli transformations**, which are basic unitary operations on a single qubit.

2.3.3 Bloch Sphere Representation

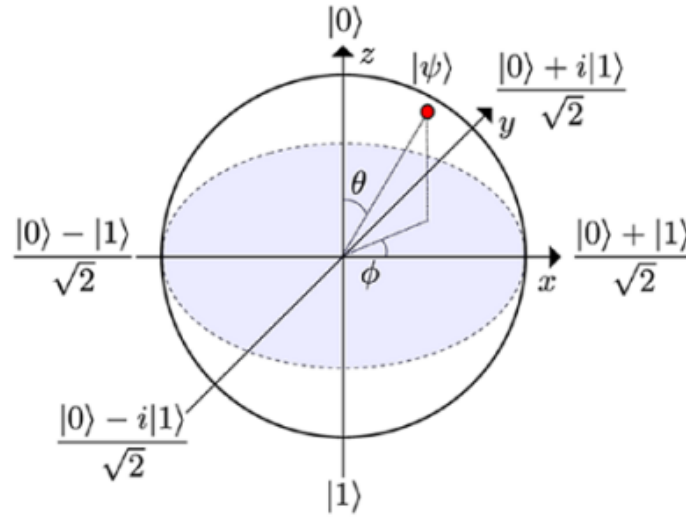


Figure 2.7: Bloch Sphere

A single qubit can be represented as:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

where $a, b \in \mathbb{C}$ and $|a|^2 + |b|^2 = 1$.

Let the complex coefficients be expressed in polar form:

$$a = a_0 + ia_1 = r_1 e^{i\theta_1} = r_1 (\cos \theta_1 + i \sin \theta_1)$$

$$b = b_0 + ib_1 = r_2 e^{i\theta_2} = r_2 (\cos \theta_2 + i \sin \theta_2)$$

with the normalization condition:

$$|r_1|^2 + |r_2|^2 = 1$$

Hence, the qubit state can be written as:

$$|\psi\rangle = r_1 e^{i\theta_1} |0\rangle + r_2 e^{i\theta_2} |1\rangle$$

$$|\psi\rangle = e^{i\theta_1} \left(r_1 |0\rangle + r_2 e^{i(\theta_2 - \theta_1)} |1\rangle \right)$$

Here, the global phase $e^{i\theta_1}$ has no observable effect and can be ignored. Therefore, a single qubit can be uniquely identified (up to global phase) by two real parameters $r_1, \theta = \theta_2 - \theta_1$.

We can represent it as:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle \quad (2.14)$$

where $\alpha \in [0, \pi]$ and $\beta \in [0, 2\pi)$.

These parameters (θ, ϕ) correspond to the coordinates of a point on the surface of the **Bloch**

Sphere, with: - θ : the polar angle (latitude), - ϕ : the azimuthal angle (longitude).

Every pure qubit state can thus be represented geometrically as a point on the unit sphere in three-dimensional space.

2.3.4 Kronecker Product (Tensor Product) of Qubits

The Kronecker (or tensor) product of two qubit state vectors is defined as:

$$\begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a \begin{pmatrix} c \\ d \end{pmatrix} \\ b \begin{pmatrix} c \\ d \end{pmatrix} \end{pmatrix} = \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix}$$

This represents the combined two-qubit system obtained from two single-qubit states.

Examples

$$|0\rangle \otimes |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$|0\rangle \otimes |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$|1\rangle \otimes |0\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$|1\rangle \otimes |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

General Form

For arbitrary qubit states,

$$\begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$$

Note: The resulting vector in general cannot be decomposed as

$$(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle)$$

unless the two qubits are separable (i.e., not entangled). For entangled states, such a factorization does not exist.

2.3.5 Two Qubit Gate

Computational Basis of Two Qubit State Space

The computational basis for a two-qubit system is given by:

$$B = \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$$

Controlled-NOT (CNOT) Gate

The matrix representation of the CNOT gate is:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

The operation of the CNOT gate on the computational basis states is as follows:

$$\text{CNOT}|00\rangle = |00\rangle$$

$$\text{CNOT}|01\rangle = |01\rangle$$

$$\text{CNOT}|10\rangle = |11\rangle$$

$$\text{CNOT}|11\rangle = |10\rangle$$

Entangled State Formation

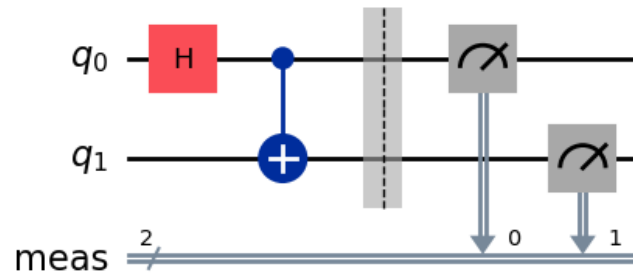


Figure 2.8: Entangled state

Let

$$|\psi\rangle = \text{CNOT}((H|0\rangle) \otimes |0\rangle)$$

Since

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle),$$

we get

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

After applying CNOT:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

which is an entangled state.

Proof of Non-Factorizability

Compare

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

with the general separable form

$$(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$$

Equating coefficients:

$$ac = \frac{1}{\sqrt{2}} \quad (i)$$

$$bd = \frac{1}{\sqrt{2}} \quad (ii)$$

$$ad = 0 \quad (iii)$$

$$bc = 0 \quad (iv)$$

From (i), $a \neq 0$ and $c \neq 0$. From (iv), either $b = 0$ or $c = 0$. Since $c \neq 0$, we must have $b = 0$. But if $b = 0$, then (ii) implies $bd = 0$, which contradicts (ii). Hence, $|\psi\rangle$ cannot be factorized and is therefore an **entangled state**.

2.3.6 n-Qubit Quantum State

An n -qubit quantum system can exist in a superposition of all 2^n possible basis states. The general state vector is given as:

$$|\psi\rangle = a_0|0\rangle + a_1|1\rangle + a_2|2\rangle + \cdots + a_{2^n-1}|2^n-1\rangle \quad (2.15)$$

where each complex coefficient a_i represents the amplitude associated with the corresponding basis state $|i\rangle$.

For example, the state $|0\rangle^{\otimes n}$ (read as “ n copies of $|0\rangle$ ”) is represented as:

$$|0\rangle^{\otimes n} = |00 \dots 0\rangle$$

In the general form, we can write:

$$|\psi\rangle = \sum_{i=0}^{2^n-1} a_i|i\rangle$$

To ensure the state is normalized, the following condition must hold:

$$\sum_{i=0}^{2^n-1} |a_i|^2 = 1 \quad (2.16)$$

2.4 Qiskit

2.4.1 Environment Setup and Qiskit Installation

To work with Qiskit, a dedicated Python virtual environment was created using the `venv` module. This ensures that all dependencies remain isolated from the system-wide Python installation.

```
[language=bash] Create a virtual environment python3 -m venv .venv
Activate the virtual environment source .venv/bin/activate
Install Qiskit pip install qiskit qiskit_aer qiskit_ibm_runtime qiskit_textbook
```

2.4.2 Quantum Circuit Creation and Simulation using Qiskit

The following Python code creates a 2-qubit quantum circuit in Qiskit. A Hadamard gate places the first qubit into superposition, and a CNOT gate entangles the second qubit with the first. Measurements are executed using the `StatevectorSampler`.

```
[language=Python] from qiskit import QuantumCircuit from qiskit.primitives import StatevectorSampler
from qiskit.visualization import plot_histogram
qc = QuantumCircuit(2) qc.h(0) qc.cx(0, 1) qc.measure_all()
qc.draw('mpl') Saves/Shows circuit diagram
sampler = StatevectorSampler() result = sampler.run([qc], shots=1024).result()
counts = result[0].data.meas.get_counts() print(counts) Out put : '00' : 504, '11' : 520
plot_histogram(counts)
```

Experiment result (circuit execution output):

```
{'00': 504, '11': 520}
```

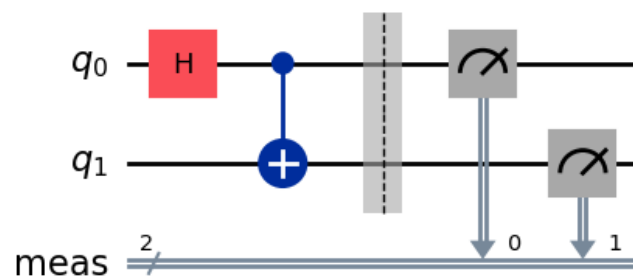


Figure 2.9: Generated quantum circuit

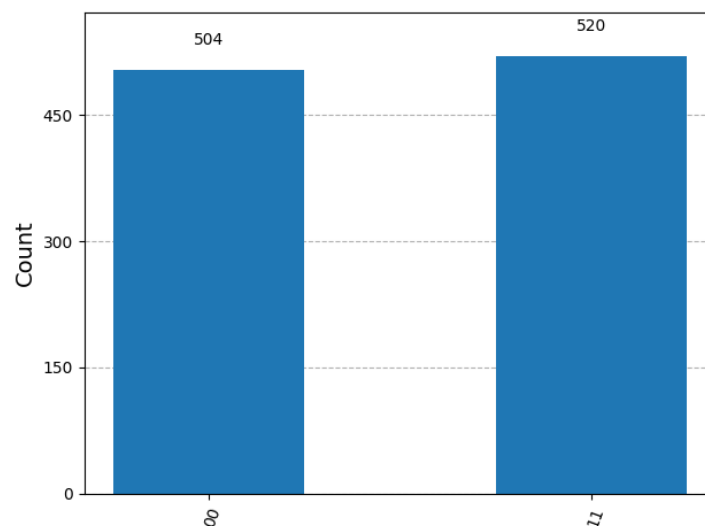


Figure 2.10: Measurement histogram (Bell state output)

2.5 Evaluation Metrics and Loss Functions

This section provides an overview and mathematical formulation of the commonly used performance metrics and loss functions for image segmentation and classification models.

2.5.1 Accuracy

Overview: Accuracy measures the proportion of correctly classified samples among all samples. It is a basic yet important indicator of model performance.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2.17)$$

where TP , TN , FP , and FN denote True Positives, True Negatives, False Positives, and False Negatives respectively.

2.5.2 Precision

Overview: Precision quantifies how many of the predicted positive samples are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2.18)$$

2.5.3 Recall (Sensitivity)

Overview: Recall measures how many of the actual positive samples are correctly predicted by the model.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2.19)$$

2.5.4 F1-Score

Overview: F1-Score is the harmonic mean of Precision and Recall, providing a balanced metric when both are important.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (2.20)$$

2.5.5 Dice Coefficient (Dice Score)

Overview: Dice Coefficient evaluates the overlap between the predicted segmentation and the ground truth mask. It ranges from 0 (no overlap) to 1 (perfect overlap).

$$\text{Dice Score} = \frac{2|P \cap G|}{|P| + |G|} = \frac{2TP}{2TP + FP + FN} \quad (2.21)$$

2.5.6 Intersection over Union (IoU)

Overview: IoU, also known as the Jaccard Index, measures the similarity between the predicted and true regions.

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|} = \frac{TP}{TP + FP + FN} \quad (2.22)$$

2.5.7 Binary Cross-Entropy Loss (BCE Loss)

Overview: BCE Loss is used for binary classification tasks, comparing the predicted probability with the ground truth label.

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \quad (2.23)$$

where y_i is the ground truth label and p_i is the predicted probability.

2.5.8 BCE with Logits Loss

Overview: BCE with Logits Loss combines a Sigmoid layer and the BCE loss in one single step for numerical stability.

$$\mathcal{L}_{\text{BCE-Logits}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\sigma(z_i)) + (1 - y_i) \log(1 - \sigma(z_i))] \quad (2.24)$$

where $\sigma(z_i) = \frac{1}{1 + e^{-z_i}}$ is the Sigmoid activation applied to the logits z_i .

2.5.9 Cross-Entropy Loss

Overview: Cross-Entropy Loss measures the difference between two probability distributions — the true labels and the predicted probabilities.

$$\mathcal{L}_{\text{CE}} = -\sum_{i=1}^C y_i \log(p_i) \quad (2.25)$$

where C is the number of classes.

2.5.10 Dice Loss

Overview: Dice Loss is derived from the Dice Coefficient and is used to optimize overlap between predicted and true segmentation.

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_{i=1}^N p_i g_i + \epsilon}{\sum_{i=1}^N p_i + \sum_{i=1}^N g_i + \epsilon} \quad (2.26)$$

where p_i and g_i are predicted and ground truth pixel values, and ϵ is a small constant for numerical stability.

2.5.11 Tversky Loss

Overview: Tversky Loss generalizes Dice Loss by introducing weights α and β to control the penalty of false positives and false negatives.

$$\mathcal{L}_{\text{Tversky}} = 1 - \frac{\sum_{i=1}^N p_i g_i + \epsilon}{\sum_{i=1}^N p_i g_i + \alpha \sum_{i=1}^N p_i (1 - g_i) + \beta \sum_{i=1}^N (1 - p_i) g_i + \epsilon} \quad (2.27)$$

where $\alpha, \beta \in [0, 1]$ and typically $\alpha = \beta = 0.5$ for the Dice case.

Chapter 3

Literature Survey

Recent developments in Quantum Machine Learning (QML) have accelerated innovations in medical image analysis, segmentation, and classification. Wang *et al.* [31] introduced a Quantum Moving Target Segmentation (QMTS) algorithm for grayscale video, integrating quantum background modeling, subtraction, and binarization circuits to achieve $O(n+q)$ complexity. This provided an exponential speedup over classical moving target segmentation methods and demonstrated feasibility on IBM Q within the NISQ framework. Similarly, Wenjie Liu *et al.* proposed a quantum image segmentation algorithm based on grayscale morphology using the Novel Enhanced Quantum Representation (NEQR) model [32]. The algorithm performs quantum morphological operations such as dilation, erosion, top-hat, and bottom-hat transformations simultaneously on all pixels, reducing computational complexity to $O(n^2 + q)$ and enabling real-time segmentation. Complementing these works, Jiale muet *et al.* developed a quantum-guided image filtering algorithm [33] under the NEQR model, reducing the time complexity from $O(2^{2q})$ to $O(q^2)$ and improving filtering quality through quantum parallelism. Mario Mastriani [34] further contributed to this domain by introducing a quantum edge detector operating on computational basis states with a projection-based Bloch Sphere logic, enhancing robustness and quantum–classical interfacing for optical image segmentation.

In the area of unsupervised and clustering-based quantum methods, Taoreed A. Akinola *et al.* presented an inverse Quantum Fourier Transform (IQFT)-inspired algorithm [35] for unsupervised image segmentation, leveraging quantum probability correlations to achieve up to 11.1% improvement in Intersection-over-Union (IoU) compared to classical k-means. Agrawal *et al.* introduced Quantum-Inspired Particle Swarm Optimization (QIPSO) and Quantum-Inspired Enhanced PSO (QIEPSO) algorithms that integrated quantum principles with metaheuristic optimization to enhance segmentation accuracy, convergence speed, and robustness [36]. Bhattacharyya *et al.* proposed a Quantum-Inspired Fractional Order Ant Colony Optimization (QIFACO) algorithm [37] for hyperspectral image clustering, combining fractional pheromone updating, PCA-based preprocessing, and K-Harmonic Means clustering to achieve superior segmentation results on the Xuzhou HYSPEX dataset. These quantum-inspired clustering and segmentation approaches collectively demonstrate how quantum principles can improve computational speed, stability, and segmentation precision beyond classical techniques.

The emergence of hybrid quantum–classical architectures has also transformed complex image

processing and classification tasks. Rao *et al.* developed GAN-Q-UNet [17], a model combining Generative Adversarial Networks with Quantum U-Net architecture for enhanced 2D–3D spine X-ray registration, which achieved superior PSNR, SSIM, and DSC values. Jain *et al.* [28] introduced Quantum Feature Extraction (QuFeX) and Qu-Net, embedding quantum modules within U-Net to improve segmentation accuracy and reduce computational complexity. Sagingalieva *et al.* proposed a quantum-inspired hyperparameter optimization framework using tensor-train optimization integrated with a hybrid ResNet model [38], achieving 0.97 accuracy and faster convergence compared to classical grid search methods. Gordienko *et al.* explored hybrid quantum–classical neural networks by integrating quanvolutional layers with ResNet, VGG16, and EfficientNet architectures [39], resulting in scalable, energy-efficient image classification across datasets such as CIFAR100 and satellite imagery. Similarly, Slabbert and Petruccione combined a ResNet10-based convolutional autoencoder for classical feature extraction with a Quantum Support Vector Machine (QSVM) classifier, effectively reducing noise and dimensionality in NISQ-based image classification.

In the biomedical imaging domain, quantum-enhanced and hybrid QML models have demonstrated remarkable diagnostic potential. Lalitha Kumari Pappala *et al.* developed a suite of quantum-enhanced biomedical image processing frameworks including Q-SVM-EFE for feature extraction [40]. Li *et al.* proposed a hybrid deep learning–quantum computing framework for lung cancer detection from CXR and CT scans, achieving 92.12% accuracy, 94% sensitivity, and 90% specificity. Zhou *et al.*

Finally, the integration of attention and transformer mechanisms with quantum approaches is shaping the next wave of medical image analysis. Zhang *et al.* [41] conducted a comprehensive review of attention mechanisms in medical image segmentation, classifying nearly 200 studies into pre-Transformer, Transformer, and Mamba-based attention categories. Their findings emphasize advancements in attention-guided architectures while identifying key challenges such as robustness, standardization, and reproducibility. Together, these works underscore the evolving synergy between quantum computing, deep learning, and attention mechanisms—marking a transition toward intelligent, scalable, and energy-efficient quantum-augmented imaging systems capable of addressing modern healthcare challenges.

Ref	Year	Technique	Author	Performance	Pros	Cons
[37].	2017	QIACO	Siddhartha Bhattacharyya <i>et al</i>		PSNR is found higher in MQIFOACO	Automatic detection techniques comes handy
[42].	2023	ML classifiers, QSVM, Pre-trained DL models, and Swin-T ROP model.	V. M. RAJA SANKARI <i>et al</i>	95.5% accuracy and a 0.97 AUC	Combines QML with ML/CNN to overcome classical limitations in ROP prediction	The paper mentions relatively higher false negatives (ROP predicted as normal), which is risky in a clinical setting.
[43].	2024	Hybrid Quantum–Classical Image Classification using Autoencoder and Quantum SVM	Donovan Slabbert <i>et al</i>		Compatible with small-scale NISQ quantum devices	The method is still being tested and is not ready to be used in real-life situations yet.
[44].	2024	Hybrid quantum–classical approach combining auto-encoders with quantum support vector machines (QSVMs).	Jason Elroy Martis <i>et al</i>		Loss Curve shows normal decrease as epochs increases	The framework was tested on a limited dataset and needs validation on larger, unseen data for standardization.
[28].	2025	QuFeX	Naman Jain <i>et al</i>	Around 94% accuracy	Compatible with small-scale NISQ quantum devices	The method is still being tested and is not ready to be used in real-life situations yet.
[32].	2022	Quantum Image Segmentation Based on Grayscale Morphology	Wenjie Li <i>et al</i>		For a 2×2 image with q grayscale levels, the algorithm's complexity reduces to $O(n^2 + q)$.	

Table 3.1: Summary of hybrid and quantum approaches in medical and image-based analysis (Ref 30-35).

Ref	Year	Technique	Author	Performance	Pros	Cons
[45].	2022	Automatic clustering of colour images using quantum inspired meta-heuristic algorithms	Alokananda Dey <i>et al</i>		QIEPSO and QIPSO demonstrate better convergence and segmentation quality compared to PSO, EPSO, ABC, DE, and CMA-ES	"Quantum inspired" algorithms do not actually run on quantum hardware. The benefits are simulated, not hardware-based
[34].		Quantum Image Processing using FRQI/NEQR-based Quantum Edge Detection	Mario Mastriani <i>et al</i>		Proposes practical ways to send data into a quantum system (classical-to-quantum) and get it back (quantum-to-classical)	The method avoids decoherence only because it uses basis states (0 or 1). In real systems with superpositions, decoherence is still a big issue
[46].	2021	INCQI	Jie Suet <i>al</i>		Suitable for future real-world quantum image applications.	Needs many quantum bits, which real quantum computers can't handle yet for big images.
[47]	2024	Medical Quantum Convolutional Neural Network (MQCNN)	Esraa Hassanet <i>al</i>	Around 99.6% accuracy (Also achieved 99.7% precision, 99.6% recall, and 99.7% F1-score)	Improved model generalization and learning capability.	Complex architecture — difficult to interpret and implement.

Table 3.2: Summary of hybrid and quantum approaches in medical and image-based analysis (Ref 36-39).

Ref	Year	Technique	Author	Performance	Pros	Cons
[48].	2023	GAN with Quantum UNet Model	Srinivasa Rao Gadu <i>et al</i>	Around 93.9% (SSIM), indicating high-quality and precise 3D image registration	The use of QNN improves the training process by storing multiple values at the same time	The real-time x-ray images are collected from the dataset. But it doesn't contain the 3D information
[31].	2024	QVNEQR	Lu Wang <i>et al</i>		Accurate and complete segmentation	High Qubit Consumptiobn
[40].	2025	Quantum Support Vector Machine with Enhanced Feature Extraction (Q-SVM-EFE)	Lalitha Kumari Pappala <i>et al</i>	Around 96.4% accurate	Enhances the speed and accuracy of diagnostics	Tested mainly on specific datasets (ADNI, TCIA), reducing generalizability
[38].	2023	Hybrid Quantum ResNet model with Tensor Train (TetraOpt) hyperparameter optimization.	Asel Sagingalieva <i>et al</i>	98.9% (achieved by the hybrid model with Grid Search after 75 iterations) & Uses shallow circuits and few qubits, making it suitable for current Noisy Intermediate-Scale Quantum (NISQ) hardware	Model performance depends heavily on precise hyperparameter tuning and search space	
[35].	2024	Quantum Fourier transform (IQFT)-inspired algorithm for unsupervised image segmentation	Taoreed A. Akinola <i>et al</i>	84%-91% with K-means clustering	Robust segmentation using probability thresholding	Accuracy drops in low-contrast images

Table 3.3: Summary of hybrid and quantum approaches in medical and image-based analysis (Ref 40-44).

Ref	Year	Technique	Author	Performance	Pros	Cons
[39].	2024	Hybrid Multimodal Quanvolutional and Convolutional Neural Network (HNN), combining a quanvolutional layer with classical CNN architectures such as ResNet, EfficientNet, and VGG-16	Yuri Gordienko <i>et al</i>	Up to 98.81% on the Hurricane Damage (HD) dataset using the EfficientNet-v2-s HNN model	Incorporates pre-trained models (ResNet, EfficientNet, VGG-16) for better generalization	HNNs may show 5-10% lower accuracy than classical models.
[33].	2025	Quantum Guided Image Filtering Algorithm based on the NEQR (Novel Enhanced Quantum Representation) model	Jiale Mue <i>et al</i>	Roughly 94% image quality preservation (SSIM = 0.94)	Demonstrates the power of quantum parallelism and entanglement	Still simulated on classical hardware, not yet fully quantum-executed
[41].	2025	Transformer and Mamba architectures — for accurate and efficient medical image segmentation	Jianpeng Zhang	90 – 98%, depending on dataset and model type	Improved generalization with fewer annotated samples	Pre-Transformer attentions have limited global context.
[49].	2025	Scalable Quantum Convolutional Neural Network (SQCNN)	Yuchen Sun <i>et al</i>	99.79% average classification accuracy on MNIST and Fashion MNIST datasets	Efficient feature extraction using quantum properties like superposition and entanglement	Quantum noise and decoherence can affect performance in real devices

Table 3.4: Summary of hybrid and quantum approaches in medical and image-based analysis (Ref 45-49).

Chapter 4

Proposed Methodology

In this work, we introduce two complementary methodologies: one for tumor classification and another for tumor segmentation followed by subtype classification on the segmented regions. The first proposed method, the Quantum-Driven Probabilistic Feature Extraction (QDPFE) framework, is designed to harness quantum encoding for extracting discriminative features from MRI data. The algorithm begins by downsampling each MRI slice and dividing it into non-overlapping patches. These patches are then transformed into quantum representations using an appropriate quantum encoding scheme. Based on a probabilistic thresholding mechanism applied to measurement outcomes, representative quantum feature vectors are extracted from each patch. Finally, all patch-wise features are aggregated to form a global feature set, which is used as input to a classical classifier for tumor categorization.

The second methodology focuses on tumor segmentation and subsequent type classification using a hybrid Quantum U-Net architecture. To prevent overfitting and enrich training variability, we apply extensive data augmentation using the Albumentations library. A multitask U-Net model is initially trained across multiple epochs to learn joint segmentation and classification tasks. Building upon this classical backbone, we then fine-tune our Quantum U-Net model for 10 and 20 epochs, enabling quantum-enhanced feature learning within the decoder pathway. This two-stage training strategy ensures improved segmentation accuracy while incorporating quantum transformations that strengthen the model's ability to distinguish between different tumor types.

4.1 QDPFE

In this Proposed work, the original images are downsampled to various resolutions using adaptive pooling, partitioned into non-overlapping patches, and then encoded through the proposed Quantum-Driven Probabilistic Feature Extraction (QDPFE) model. The extracted quantum-based features are subsequently classified using multiple machine learning classifiers. The overall methodology is summarized in Fig. 4.

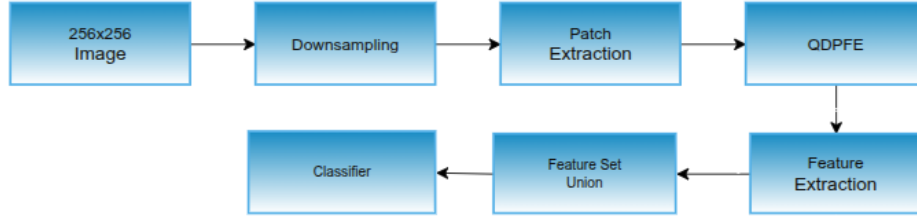


Figure 4.1: Flow Diagram of the Proposed QDPFE Approach

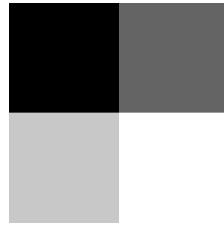
4.1.1 Downsampling

The original 256×256 grayscale images are downsampled using adaptive average pooling and adaptive max pooling techniques, producing resized versions of 16×16 , 32×32 , and 64×64 pixels for subsequent processing. These sizes ($N = 16, 32, 64$) are selected to balance computational feasibility with feature granularity rather than via systematic tuning.

4.1.2 Patch Extraction

From each downsampled image, non-overlapping patches of size $p \times p$ ($p = 2, 4$) are extracted by dividing the image into uniform grids across rows and columns. The grayscale patch extraction process is shown in Fig. 2. Each extracted patch is then forwarded to the quantum encoding stage for feature representation.

4.1.3 Quantum-Driven Probabilistic Feature Encoding

Figure 4.2: Grayscale 2×2 image used for QDPF encoding.

The Quantum-Driven Probabilistic Feature Extraction (QDPFE) model stores the grayscale intensity (C) and positional information (Y, X) of a classical image into three qubit sequences. The grayscale and positional qubits are entangled to form a unified quantum image representation.

For a classical image of size $2^n \times 2^n$ and a grayscale range $[0, 2^q - 1]$, $2n$ qubits are used to store the positional information, while q qubits encode the grayscale values. The QDPFE quantum image state is represented as:

$$|I\rangle = \frac{1}{2^n} \sum_{Y=0}^{2^n-1} \sum_{X=0}^{2^n-1} |C_{Y,X}\rangle \otimes |Y,X\rangle \quad (4.1)$$

Here, $|C_{Y,X}\rangle$ encodes the q -bit grayscale value at pixel coordinates (Y,X) , while $|Y,X\rangle$ encodes the $2n$ -bit positional index.

Example: For a 2×2 grayscale image:

$$\begin{bmatrix} 0 & 100 \\ 200 & 255 \end{bmatrix}$$

we have $n = 1$ (image size $2^n \times 2^n$) and $q = 8$ (8-bit grayscale). Binary grayscale values:

$$0 \rightarrow 00000000, \quad 100 \rightarrow 01100100, \quad 200 \rightarrow 11001000, \quad 255 \rightarrow 11111111$$

Pixel positions:

$$(0,0) \rightarrow |00\rangle, \quad (0,1) \rightarrow |01\rangle, \quad (1,0) \rightarrow |10\rangle, \quad (1,1) \rightarrow |11\rangle$$

Thus, the corresponding QDPFE state is:

$$|I\rangle = \frac{1}{2} (|00000000\rangle|00\rangle + |01100100\rangle|01\rangle + |11001000\rangle|10\rangle + |11111111\rangle|11\rangle)$$

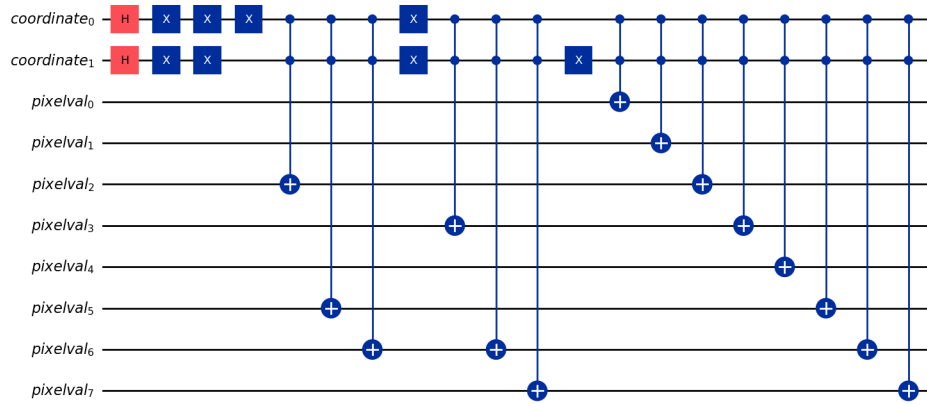


Figure 4.3: QDPFE quantum circuit representation of the 2×2 image in Fig. 4.2.

4.1.4 Probability-Based Feature Extraction

For each extracted image patch, the grayscale and spatial coordinates are encoded into a quantum circuit based on the QDPFE representation. A dictionary is constructed to map each pixel's intensity and coordinates (y,x) along with the number of qubits used for vertical (q_y) and horizontal (q_x) positioning.

After constructing the quantum circuit (shown in Fig. 3), all qubits are measured simultaneously for a fixed number of shots (e.g., 1000). The resulting measurement counts are converted into

normalized probabilities. A statistical threshold T is then computed as:

$$T = \mu_p + k \cdot \sigma_p \quad (4.2)$$

where μ_p and σ_p denote the mean and standard deviation of the probabilities, respectively, and k is a tunable scaling factor. Basis states with probabilities greater than T are retained as significant features.

4.1.5 Feature Set Union

The feature sets obtained from all patches of a single image are unified to form the complete feature set using the union operation:

$$F_{\text{final}} = F_{\text{patch}_1} \cup F_{\text{patch}_2} \cup \dots \cup F_{\text{patch}_n} \quad (4.3)$$

This ensures that all discriminative quantum-based features are aggregated for downstream analysis.

4.1.6 Classical Classifiers

The final extracted feature set is used for classification using multiple classical machine learning models:

- **Support Vector Machine (SVM):** Tested with linear and RBF kernels.
- **Logistic Regression:** Applied with a maximum iteration limit of 500.
- **K-Nearest Neighbors (KNN):** Evaluated for neighbors $k = 3, 5, 7$ using both uniform and distance-based weights.
- **Random Forest:** Tested with estimators = 50, 100, 200 and max depth = None, 10, 20.
- **Gradient Boosting:** Evaluated with estimators = 50, 100, 200 and learning rates = 0.01, 0.1, 0.2.

All classifiers were tuned on their respective hyperparameters, and only the best-performing configurations were considered for evaluation. The overall model pipeline is illustrated in Fig. 4.

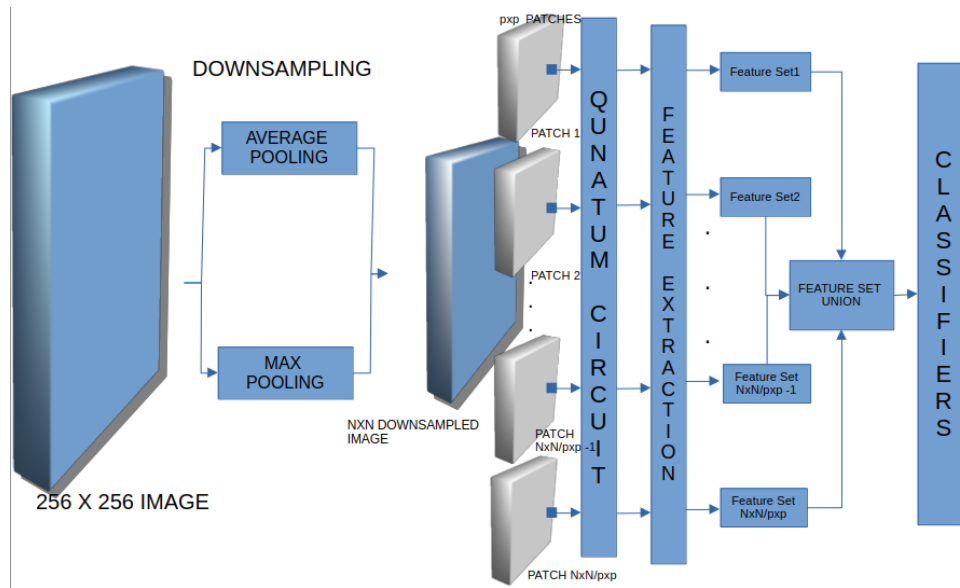


Figure 4.4: Overall model of the proposed feature extraction and classification approach.

4.1.7 Algorithm QDPFE

Algorithm 1: Quantum-Driven Probabilistic Feature Extraction (QDPFE) and Classification Pipeline

Original grayscale image I_{orig} of size 256×256 ;

Downsample sizes $N = \{16, 32, 64\}$;

Patch sizes $p = \{2, 4\}$;

Grayscale bit-depth q (e.g., $q = 8$);

Number of quantum shots $S = 1000$;

Statistical parameter k for thresholding.

Final feature set F_{final} and classification results.

Step 1: Downsampling N in $\{16, 32, 64\}$ Apply **Adaptive Average Pooling** and **Adaptive Max Pooling** to I_{orig} Obtain resized image $I_{resized}$ of size $N \times N$

Step 2: Patch Extraction $I_{resized}$ Divide $I_{resized}$ into non-overlapping patches of size $p \times p$ ($p \in \{2, 4\}$) Store all patches $\{P_1, P_2, \dots, P_n\}$ for further encoding

Step 3: Quantum-Driven Probabilistic Feature Encoding (QDPFE) patch P_i Extract grayscale values C and spatial coordinates (Y, X) Map grayscale (C) and position (Y, X) to quantum qubits: $2n$ qubits $\rightarrow$ position encoding (Y, X) q qubits $\rightarrow$ grayscale encoding Construct quantum image state:

$$|I\rangle = \frac{1}{2^n} \sum_{Y=0}^{2^n-1} \sum_{X=0}^{2^n-1} |C_{YX}\rangle \otimes |YX\rangle$$

Execute the quantum circuit on a simulator backend for S shots Obtain measurement counts and normalize into probabilities p_i

Step 4: Probability-Based Feature Extraction Compute mean μ_p and standard deviation σ_p of probability distribution Calculate threshold:

$$T = \mu_p + k \cdot \sigma_p$$

Select feature states with probability $p_i > T$ to form feature set F_{patch_i}

Step 5: Feature Set Union Combine all patch feature sets:

$$F_{final} = \bigcup_{i=1}^n F_{patch_i}$$

Step 6: Classical Classification classifier $\in \{\text{SVM, Logistic Regression, KNN, Random Forest, Gradient Boosting}\}$ Train classifier on F_{final} using tuned parameters Evaluate using accuracy, precision, recall, and F1-score Save best-performing model checkpoint

return F_{final} and best classification results

4.2 Methodology

4.2.1 Data Augmentation

During the training phase, a comprehensive data augmentation strategy was implemented using the Albumentations library to enhance model robustness, improve generalization capability, and mitigate overfitting—an essential consideration in medical image segmentation tasks where annotated datasets are typically limited. All input images were first resized to a fixed spatial resolution of 256×256 to ensure dimensional consistency and stable batch-wise optimization. To promote spatial invariance and enable the model to learn orientation-independent feature representations, random horizontal and vertical flips, along with rotations in multiples of 90° , were applied. These transformations encourage the network to recognize anatomical structures irrespective of orientation, which is particularly relevant for cross-sectional MRI and CT imaging modalities. To simulate realistic anatomical variability and strengthen robustness to geometric distortions, non-linear spatial transformations were incorporated, including smooth elastic deformations and structured grid-based distortions that approximate natural tissue elasticity and minor acquisition-induced irregularities. Additionally, controlled affine transformations involving small translations, scaling variations, and angular rotations were employed to prevent the model from overfitting to rigid spatial configurations. To address inter-scanner and inter-protocol intensity variability commonly observed in clinical practice, photometric augmentations were introduced. Random adjustments to brightness and contrast, as well as mild blurring effects, were applied to emulate imaging noise and subtle acquisition artifacts. These intensity perturbations encourage the model to learn stable feature representations that are not overly sensitive to specific intensity distributions. Finally, input images were normalized using standardized mean and standard deviation values to stabilize gradient propagation and facilitate efficient convergence during training, and subsequently converted into tensor format for GPU-accelerated computation. During validation, augmentation was restricted to deterministic preprocessing steps—namely resizing, normalization, and tensor conversion—to ensure that evaluation metrics accurately reflect the true generalization performance of the trained model. Collectively, this augmentation framework serves as a principled regularization strategy that promotes the learning of anatomically meaningful and transformation-invariant representations suitable for reliable clinical deployment.

4.2.2 Multi-Task U-Net Architecture

The proposed model is a multi-task deep learning architecture based on the U-Net framework [ronneberg], designed to perform both pixel-level segmentation and image-level classification simultaneously from a single input image. The architecture follows the classical encoder–decoder paradigm of U-Net, where the encoder (contracting path) is responsible for hierarchical feature extraction and the decoder (expanding path) reconstructs high-resolution spatial representations for precise segmentation. In addition to the segmentation pathway, a dedicated classification head is introduced at the bottleneck level, enabling the network to learn global semantic representations

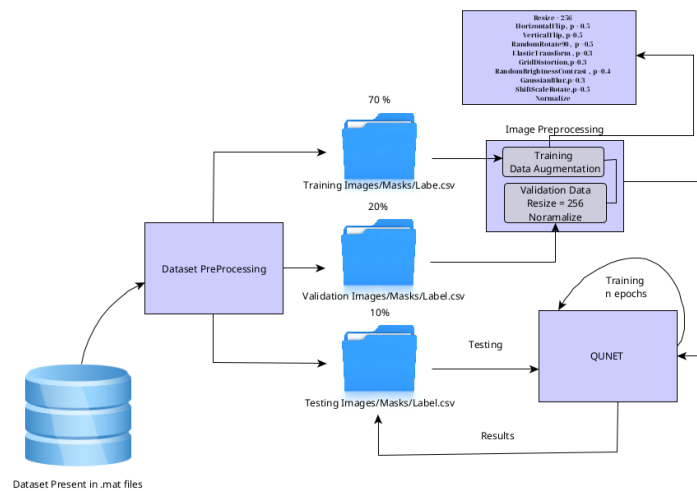


Figure 4.5: Multitasking UNET

useful for classification tasks. This shared-feature multi-task design allows the model to efficiently exploit common representations while optimizing task-specific outputs. At the core of the network lies the DoubleConv module, which serves as the fundamental building block throughout both the encoder and decoder paths. Each DoubleConv block consists of two consecutive convolutional layers with a kernel size of 3×3 and padding configured to preserve spatial dimensions. Each convolution is followed by batch normalization and a ReLU activation function, ensuring stable training and effective non-linear feature learning. A dropout layers are incorporated to provide regularization and reduce overfitting, particularly in the early encoder stages where low-level features are learned. This repeated convolutional structure enables the network to progressively refine feature representations at each resolution level. The encoder comprises five stages, beginning with an initial convolution block that transforms the input image channels into a higher-dimensional feature space. This is followed by four downsampling stages implemented using the Down module, each consisting of a 2×2 max-pooling operation followed by a DoubleConv block. The max-pooling layers systematically reduce the spatial resolution while increasing the number of feature channels, allowing the network to capture increasingly abstract and context-rich representations. The final encoder stage serves as the bottleneck of the network, where the deepest and most compressed feature maps are generated. To enhance generalization, strong dropout regularization is applied at this bottleneck layer, as it contains highly discriminative features that are critical for both segmentation and classification tasks. The decoder path is constructed using four Up modules, each responsible for progressively restoring the spatial resolution of the feature maps. Upsampling is performed using bilinear interpolation, which is computationally efficient and helps to avoid checkerboard artifacts commonly associated with transposed convolutions. After upsampling, feature maps from the corresponding encoder layers are concatenated via skip connections, ensuring that fine-grained spatial details lost during downsampling are reintroduced into the decoding process. These concatenated feature maps are then refined using DoubleConv blocks, allowing the network to effectively integrate low-level spatial information with high-level semantic context.

For the segmentation task, the decoder output is passed through a dedicated segmentation head implemented using a 1×1 convolution (OutConv). This layer maps the final feature representation to the desired number of segmentation classes, producing dense, pixel-wise predictions. The use of a 1×1 convolution ensures that spatial resolution is preserved while enabling efficient channel-wise classification. This design is particularly well suited for medical image segmentation tasks, where accurate boundary delineation and precise spatial localization are of critical importance. In parallel with the segmentation pathway, a classification head is attached directly to the bottleneck features, enabling the model to perform image-level classification. The classification branch begins with an adaptive average pooling layer that compresses the spatial dimensions to a fixed 1×1 representation, effectively aggregating global contextual information from the feature maps. This is followed by fully connected layers with ReLU activation and dropout regularization, culminating in a final linear layer that outputs class logits. By operating on bottleneck-level features, the classification head leverages the most abstract and semantically rich representations learned by the encoder.

During forward propagation, the input image is first processed through the shared encoder and bottleneck layers. Subsequently, the network branches into two task-specific pathways: the segmentation branch reconstructs high-resolution spatial predictions through the decoder and skip connections, while the classification branch performs global inference based on compressed bottleneck features. This joint learning strategy enables the network to capture both local and global information efficiently, making the architecture particularly suitable for complex medical imaging applications that require accurate localization alongside holistic interpretation.

Encoder

- I. CONV2D 1: output: $128 \times 128 \times 64$, kernel size: 3×3 , activation function: 'ReLU';
- II. CONV2D 2: output: $128 \times 128 \times 64$, kernel size: 3×3 , activation function: 'ReLU';
- III. DROPOUT: rate: 0.1;
- IV. MAX_POOLING2D: output: $64 \times 64 \times 64$, pool size: 2×2 ;
- V. CONV2D 3: output: $64 \times 64 \times 128$, kernel size: 3×3 , activation function: 'ReLU';
- VI. CONV2D 4: output: $64 \times 64 \times 128$, kernel size: 3×3 , activation function: 'ReLU';
- VII. MAX_POOLING2D: output: $32 \times 32 \times 128$, pool size: 2×2 ;
- VIII. CONV2D 5: output: $32 \times 32 \times 256$, kernel size: 3×3 , activation function: 'ReLU';
- IX. CONV2D 6: output: $32 \times 32 \times 256$, kernel size: 3×3 , activation function: 'ReLU';
- X. MAX_POOLING2D: output: $16 \times 16 \times 256$, pool size: 2×2 ;
- XI. CONV2D 7: output: $16 \times 16 \times 512$, kernel size: 3×3 , activation function: 'ReLU';
- XII. CONV2D 8: output: $16 \times 16 \times 512$, kernel size: 3×3 , activation function: 'ReLU';
- XIII. MAX_POOLING2D: output: $8 \times 8 \times 512$, pool size: 2×2 ;
- XIV. CONV2D 9: output: $8 \times 8 \times 512$, kernel size: 3×3 , activation function: 'ReLU';
- XV. CONV2D 10: output: $8 \times 8 \times 512$, kernel size: 3×3 , activation function: 'ReLU';
- XVI. DROPOUT: rate: 0.5;

BOTTLENECK LAYER

XVII. A GLOBAL AVERAGE POOLING: output: $1 \times 1 \times 512$;
 XVII. B FLATTEN;
 XVII. C DENSE: units: 256, activation function: 'ReLU';
 XVII. D DENSE: units: 256, activation function: 'ReLU';
 XVII. E DROPOUT: rate: 0.5;
 XVII. F DENSE: units: 3 (classification output);
 XVII. E QUANTUM BOTTLENECK: input: $8 \times 8 \times 512 \rightarrow \text{AdaptiveAvgPool} \rightarrow$
 Fully Connected ($512 \rightarrow 4$) $\rightarrow$ Quantum Layer (4 qubits) $\rightarrow$ Fully Connected ($4 \rightarrow 512$);

Decoder

XVIII. UP_SAMPLING2D: output: $16 \times 16 \times 512$, kernel size: 2×2 ;
 XIX. CONCATENATE: output: $16 \times 16 \times 1024$;
 XX. CONV2D 11: output: $16 \times 16 \times 256$, kernel size: 3×3 , activation function: 'ReLU';
 XX. CONV2D 11: output: $16 \times 16 \times 256$, kernel size: 3×3 , activation function: 'ReLU';
 XXI. CONV2D 12: output: $16 \times 16 \times 256$, kernel size: 3×3 , activation function: 'ReLU';
 XXII. UP_SAMPLING2D: output: $32 \times 32 \times 256$, kernel size: 2×2 ;
 XXIII. CONCATENATE: output: $32 \times 32 \times 512$;
 XXIV. CONV2D 13: output: $32 \times 32 \times 128$, kernel size: 3×3 , activation function: 'ReLU';
 XXV. CONV2D 14: output: $32 \times 32 \times 128$, kernel size: 3×3 , activation function: 'ReLU';
 XXVI. UP_SAMPLING2D: output: $64 \times 64 \times 128$, kernel size: 2×2 ;
 XXVII. CONCATENATE: output: $64 \times 64 \times 256$;
 XXVIII. CONV2D 15: output: $64 \times 64 \times 64$, kernel size: 3×3 , activation function: 'ReLU';
 XXIX. CONV2D 16: output: $64 \times 64 \times 64$, kernel size: 3×3 , activation function: 'ReLU';
 XXX. UP_SAMPLING2D: output: $128 \times 128 \times 64$, kernel size: 2×2 ;
 XXXI. CONCATENATE: output: $128 \times 128 \times 128$;
 XXXII. CONV2D 17: output: $128 \times 128 \times 64$, kernel size: 3×3 , activation function: 'ReLU';
 XXXIII. CONV2D 18: output: $128 \times 128 \times 64$, kernel size: 3×3 , activation function: 'ReLU';
 XXXIV. CONV2D (Output – Segmentation): output: $128 \times 128 \times 1$, kernel size: 1×1 ;

4.2.3 Quantum Bottleneck

The proposed implementation introduces a hybrid quantum–classical bottleneck module that integrates a parameterized quantum circuit into a deep learning architecture using Qiskit and PyTorch. The design follows the principles of Quantum Neural Networks (QNNs), wherein classical data is encoded into a quantum state, processed through a variational quantum circuit, and subsequently mapped back into the classical feature space. This bottleneck is particularly suitable for insertion at the deepest layer of convolutional neural networks, such as U-Net, where feature representations are compact yet semantically rich. By combining classical neural layers with quantum computation, the architecture aims to enhance representational expressiveness

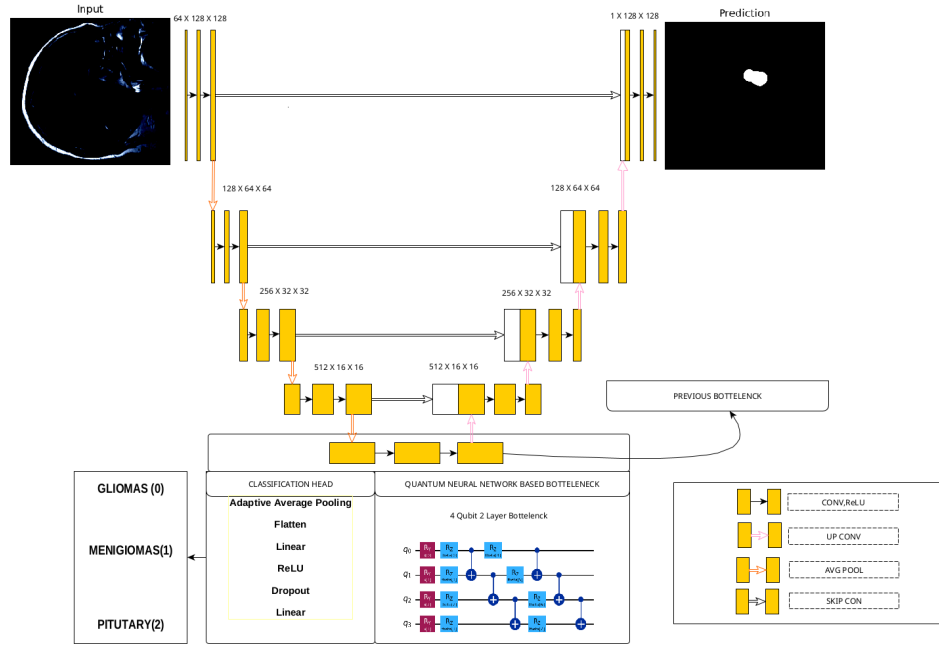


Figure 4.6: Quantum UNET

while maintaining compatibility with gradient-based optimization. The Algorithm `build_qiskit_torch_layer` constructs the core quantum neural network using a parameterized quantum circuit. A fixed number of qubits is defined, where each qubit corresponds to one dimension of the compressed classical feature vector. Classical inputs are encoded into the quantum state using rotation gates around the Y-axis (R_y), which implement an angle-encoding strategy to map real-valued features into quantum amplitudes. Trainable parameters are introduced through variational rotation gates around the Z-axis (R_z), which act as learnable weights of the quantum model. To enable the modeling of complex correlations between features, a chain of controlled-NOT (CNOT) gates is applied to entangle neighboring qubits, allowing the circuit to capture higher-order feature interactions.

The quantum circuit is executed using a statevector-based sampler, providing an exact simulation of the quantum state without hardware noise. The circuit output is interpreted as a probability distribution over all computational basis states, resulting in an output dimension of 2^n , where n denotes the number of qubits. The `SamplerQNN` abstraction is employed to define the quantum neural network by explicitly specifying the input parameters, trainable weight parameters, and output interpretation. The quantum model is then wrapped for integration with the PyTorch computational graph and allowing end-to-end training via backpropagation. During the forward pass, incoming convolutional feature maps are first spatially compressed using adaptive average pooling, reducing the spatial dimensions to 1×1 while preserving channel-wise information. Subsequently, a linear projection layer reduces the channel dimension to match the number of qubits, ensuring compatibility between the classical features and the limited input capacity of the quantum circuit. Prior to quantum processing, the compressed features are normalized using a hyperbolic tangent activation function, constraining the input range and improving

numerical stability for quantum gate operations. The quantum layer produces a probability distribution corresponding to all possible measurement outcomes. To obtain a compact classical representation, this distribution is transformed into an expectation-based bit vector, where each component represents the expected value of a specific qubit. This operation effectively maps the high-dimensional quantum output back into a low-dimensional classical feature vector. A final linear projection layer then expands this vector back to the original channel dimension, reconstructing a feature representation that can be seamlessly reintegrated into the classical network. The reconstructed tensor preserves the batch and channel dimensions, ensuring architectural compatibility with subsequent convolutional or decoding stages.

Overall, the hybrid quantum bottleneck functions as a learnable non-linear transformation that augments classical deep learning with quantum computational principles. By embedding a variational quantum circuit within a neural network, the model leverages quantum properties such as superposition and entanglement to enrich feature representations. At the same time, the architecture remains fully differentiable and trainable using standard optimization techniques, making it a practical and scalable approach for investigating quantum-enhanced learning in applications involving high-dimensional data, such as medical image analysis.

Algorithm 2: build_qiskit_torch_layer

Number of qubits n , number of variational layers L Torch-compatible quantum layer and output dimension 2^n Initialize input parameter vector $\mathbf{x} = \{x_1, x_2, \dots, x_n\}$ Initialize trainable parameter vector $\theta = \{\theta_1, \theta_2, \dots, \theta_{nL}\}$ Create a quantum circuit with n qubits $i = 1$ to n Apply angle encoding using rotation gate:

$$R_y(x_i) \text{ on qubit } i$$

each variational layer $l = 1$ to L $q = 1$ to n Apply trainable rotation:

$$R_z(\theta_{l,q}) \text{ on qubit } q$$

$q = 1$ to $n - 1$ Apply entanglement gate:

$$\text{CNOT}(q, q + 1)$$

Define identity interpretation function for measurement outcomes Initialize a statevector-based quantum sampler Construct SamplerQNN using the circuit, input parameters, and trainable parameters Wrap the quantum neural network using TorchConnector Set output dimension to 2^n Torch quantum layer and output dimension 2^n

4.2.4 Quantum Unet

The QuantumUnet model is a quantum-enhanced Bottleneck architecture derived from a pre-trained multi-task U-Net model. The primary objective of this design is to incorporate a quantum bottleneck into an already trained classical U-Net while preserving its learned representations. By reusing the encoder, decoder, and task-specific heads of a base U-Net model, this approach

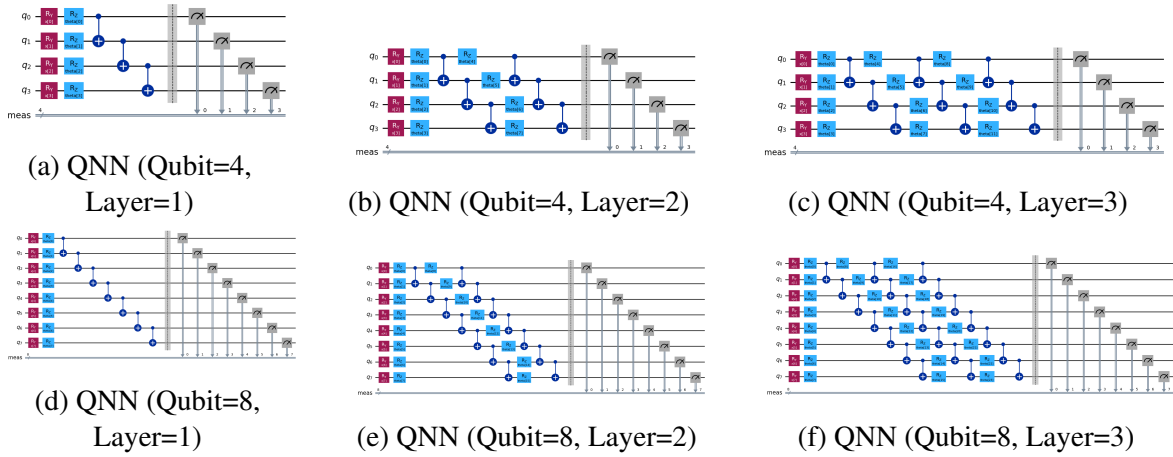


Figure 4.7: Quantum Neural Network (QNN) circuit structures for 4 and 8 qubits across 1, 2, and 3 layers.

enables efficient transfer learning and allows the investigation of quantum augmentation at the deepest feature level without retraining the entire network from scratch.

The architecture begins by inheriting the encoder components from the base U-Net model, including the initial convolution block (inc) and the four downsampling blocks (down1–down4). These layers constitute the contracting path of the U-Net and are responsible for hierarchical feature extraction at progressively coarser spatial resolutions. Similarly, the decoder blocks (up1–up4) and the segmentation output layer (out_seg) are directly reused, ensuring that the spatial reconstruction pathway remains consistent with the original trained model. In addition, the classification head is also transferred intact, enabling simultaneous segmentation and image-level classification in a multi-task learning setting and to stabilize training and preserve previously learned representations, the encoder parameters can optionally be frozen during optimization.

A central component of the proposed architecture is the insertion of a quantum bottleneck at the deepest encoder level. The number of input channels to the bottleneck is determined by the channel width of the final encoder block, adjusted by a scaling factor consistent with bilinear upsampling in the original U-Net design. Quantum Bottleneck compresses classical feature maps, processes them using a parameterized quantum circuit, and reconstructs quantum-enhanced features back into the classical domain. The quantum bottleneck is parameterized by the number of qubits and the number of variational layers, allowing flexible control over the expressiveness of the quantum circuit. During the forward pass, the input image is propagated through the shared encoder, producing intermediate feature maps at multiple resolutions. The deepest feature map is then passed through the quantum bottleneck, yielding a quantum-processed feature representation. This output is combined with the original bottleneck features using a residual connection, ensuring stable gradient flow and preserving classical information while injecting quantum-induced feature transformations. If quantum processing is disabled, a classical

dropout layer is applied instead, serving as a baseline regularization mechanism for comparison against the quantum-enhanced variant.

Following bottleneck processing, the decoder reconstructs high-resolution feature maps through successive upsampling operations and skip connections with the encoder feature maps. These skip connections allow the decoder to recover fine-grained spatial details lost during downsampling. The reconstructed feature map is passed through the segmentation head, which produces dense, pixel-wise predictions suitable for medical image segmentation tasks. In parallel, the classification head operates directly on the bottleneck feature representation, enabling global image-level prediction based on high-level semantic information.

Overall, the architecture provides a principled framework for integrating quantum computation into a transfer learning paradigm. By localizing quantum processing to the bottleneck layer and reusing a pre-trained classical backbone, the model achieves computational efficiency while enabling a controlled study of quantum feature augmentation. The dual-task design further enhances the utility of the architecture in medical imaging applications, where both precise localization and holistic classification are essential. This approach represents a scalable and experimentally feasible pathway toward hybrid quantum–classical deep learning systems.

Chapter 5

Results and Discussion

5.1 Environment Setup

The experimental framework for both the Quantum-Driven Probabilistic Feature Extraction (QDPFE) and Quantum U-Net models was established through a hybrid cloud-based and local computing environment. The configuration ensures seamless quantum-classical integration, reproducibility, and efficient GPU utilization.

5.1.1 IBM Quantum Platform Integration

To execute and simulate quantum components, the **IBM Quantum Platform** was utilized. The setup involved:

- Creating an IBM Quantum account and obtaining a personal API token.
- Logging into the IBM Quantum service using the Qiskit library through the command:

```
from qiskit_ibm_runtime import QiskitRuntimeService
service = QiskitRuntimeService(channel = "ibm_cloud",
                                token = "API_TOKEN")
QiskitRuntimeService.save_account(channel = "ibm_cloud",
                                token = "API_TOKEN")
```
- Accessing the available quantum backends (*e.g.*, `ibmq_qasm_simulator`, `ibmq_manila`) to run the Quantum circuits.
- Ensuring the use of the `AerSimulator` backend during initial simulation and debugging phases.

5.1.2 Project Directory and Drive Configuration

To maintain separation between the classical and quantum experiments, two dedicated Google Drive folders were created:

- **QDPFE Drive:** Contains classical preprocessing scripts, NEQR encoding modules, quantum feature extraction notebooks, and classifier models.
- **Quantum U-Net Drive:** Stores U-Net checkpoints, hybrid model architecture files, and training logs for multitask segmentation and classification.

Both drives were mounted in Google Colab to ensure automatic synchronization and persistent storage across sessions. The following command was used:

```
from google.colab import drive
drive.mount('/content/drive')
```

5.1.3 Google Colab GPU Configuration

The experiments were executed on Google Colab Pro, with access to an NVIDIA T4 GPU to accelerate the classical model training stages. The runtime environment was configured as follows:

- Runtime Type: GPU (T4)
- Python Version: 3.10+
- Key Libraries: PyTorch, TensorFlow, Qiskit, Scikit-learn, Matplotlib, and NumPy
- All critical files (checkpoints, plots, and logs) were automatically backed up to the linked Google Drive directories.

5.1.4 Version Control and Backup

Version tracking was ensured by saving each notebook iteration on Google Drive and maintaining consistent naming conventions for checkpoints. Regular synchronization between local and cloud copies of both QDPFE and Quantum U-Net workspaces guaranteed the reproducibility of all experiments.

5.2 Dataset Description

The experimental analysis in this study was performed using two distinct datasets: the **Br35H: Brain Tumor Detection 2020** dataset and a **FigShare medical imaging dataset**. Both datasets were used to train and evaluate the proposed classical and quantum-enhanced U-Net models under controlled preprocessing and splitting protocols.

5.2.1 Br35H Dataset

The *Br35H: Brain Tumor Detection 2020* dataset [50], hosted on **Kaggle**, consists of magnetic resonance imaging (MRI) scans of the human brain, labeled as either tumor-positive (“yes”) or tumor-negative (“no”). For computational feasibility while maintaining class balance, a subset of **300 images** was randomly selected — **150 samples per class**.

All images were converted to grayscale and uniformly resized to **256×256** pixels to ensure compatibility with the model’s input dimensions. The dataset provides a clear binary classification framework for evaluating tumor detection capabilities in medical image segmentation and classification models.

- **Source:** Kaggle (Br35H Brain Tumor Detection 2020)
- **Classes:** Tumor-Positive (“Yes”), Tumor-Negative (“No”)
- **Selected Samples:** 300 (150 per class)
- **Preprocessing:** Grayscale conversion and 256×256 resizing

5.2.2 FigShare Dataset

A second dataset was obtained from **FigShare.com** [51] in .mat file format, containing MRI image data and corresponding segmentation masks. The preprocessing pipeline involved extracting individual image slices, associated masks, and label metadata from the MATLAB files.

Following extraction, three separate folders were organized:

- `images/` — containing the input MRI slices.
- `masks/` — containing the ground-truth segmentation masks.
- `labels.csv` — containing the corresponding class labels and sample indices.

To ensure effective training and validation, the dataset was split into three subsets with the following proportions:

- **Training Set:** 70%
- **Validation Set:** 20%
- **Testing Set:** 10%

During preprocessing, care was taken to maintain a balanced class distribution and prevent data leakage between subsets. All extracted images and masks were verified for consistency and alignment to avoid any loss or mismatch in data samples.

- **Source:** FigShare.com (MATLAB-based MRI dataset)

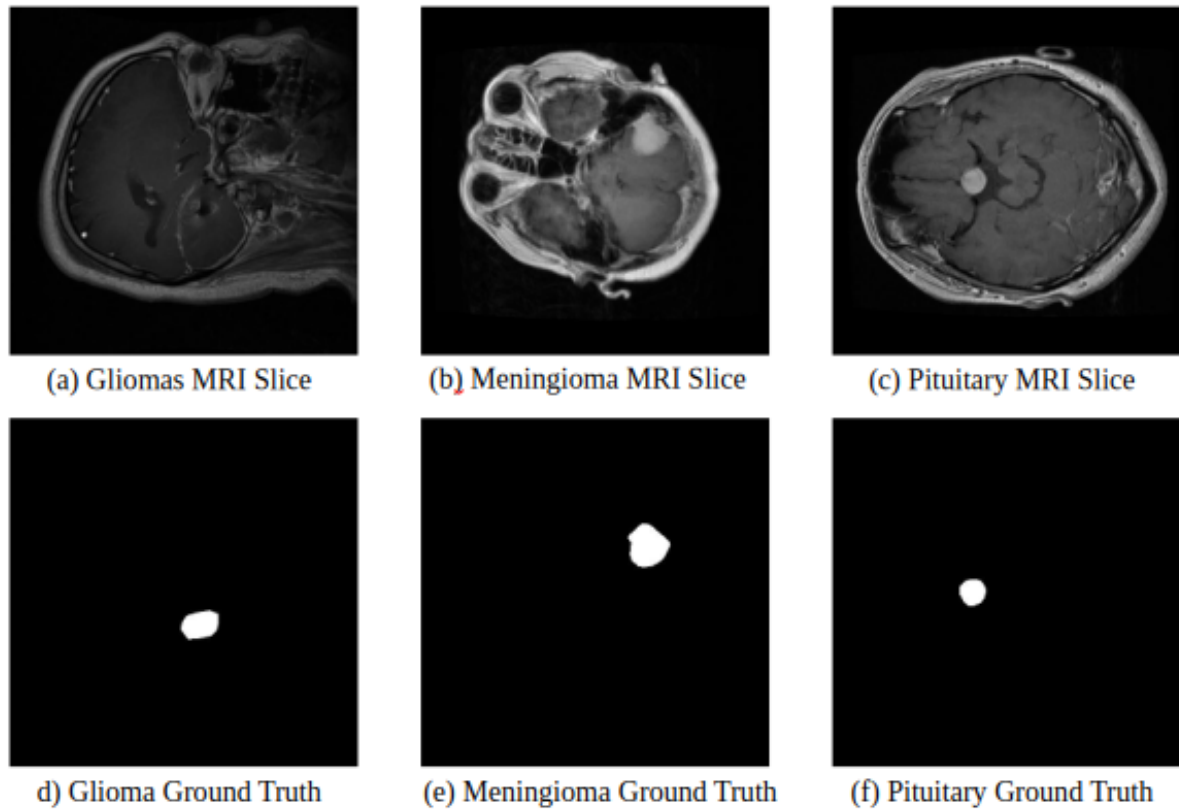


Figure 5.1: Representative MRI slices and corresponding ground-truth segmentation masks for the three tumor classes: glioma, meningioma, and pituitary. The images illustrate inter-class morphological variations and segmentation complexity.

- **File Format:** .mat (converted to image-mask pairs)
- **Data Split:** 70% Train, 20% Validation, 10% Test
- **Output Folders:** images/, masks/, labels.csv

Both datasets collectively provided the foundation for classical, multitask, and quantum-enhanced U-Net model evaluations, ensuring robust generalization across different data modalities and segmentation complexities.

5.3 Evaluation Metrics

To comprehensively assess the performance of both classical and quantum-enhanced models, multiple evaluation metrics were employed across the two primary experimental pipelines: **Quantum-Driven Probabilistic Feature Extraction (QDPFE)** and the **Quantum U-Net (QUNet)** model. Each metric was selected to capture distinct aspects of classification and segmentation accuracy, precision, and reliability.

5.3.1 Metrics for QDPFE

For the QDPFE-based feature extraction and classical classification experiments, the following standard classification metrics were utilized:

- **Accuracy:** Measures the proportion of correctly classified samples over the total number of samples.
- **Precision:** Indicates the proportion of true positive predictions among all positive predictions made by the model.
- **Recall (Sensitivity):** Represents the proportion of actual positive samples correctly identified by the classifier.
- **F1-Score:** The harmonic mean of precision and recall, providing a balanced measure between them.

These metrics were computed for each classifier used after QDPFE-based feature extraction (SVM, Logistic Regression, KNN, Random Forest, and Gradient Boosting), enabling a fair comparison of performance across different models.

5.3.2 Metrics for Quantum U-Net

For the Quantum U-Net (QUNet) and Multitask U-Net architectures, evaluation was conducted using segmentation and classification performance metrics, focusing on pixel-wise accuracy and overall classification reliability.

Segmentation Metrics:

- **Mean Dice Score:** Quantifies the overlap between the predicted and ground-truth segmentation masks.
- **Mean Dice Loss:** Computed as $1 - \text{Mean Dice Score}$, reflecting segmentation error.
- **Precision:** Evaluates the correctness of predicted positive pixels relative to all predicted positives.
- **Recall:** Measures the ability of the model to detect all relevant pixels belonging to the target class.
- **F1-Score:** Provides a combined measure of segmentation precision and recall.

Classification Metrics: For the classification head of the Quantum U-Net model, the same set of metrics—**accuracy, precision, recall, and F1-score**—were applied to evaluate categorical predictions based on image-level tumor classification.

5.3.3 Summary

Together, these metrics provide a comprehensive evaluation framework: accuracy-oriented measures ensure reliable classification, while Dice-based measures capture fine-grained spatial correspondence between predicted and ground-truth regions, essential for medical image segmentation tasks.

5.4 Experimental Setup and Analysis for QDPFE

5.4.1 Experimental Setup

In the experimental setup, the original 256×256 grayscale images are first downsampled to multiple resolutions (16×16 , 32×32 , and 64×64) using **adaptive average pooling** and **adaptive max pooling** techniques. This multi-scale downsampling ensures that both fine-grained and coarse-level spatial features are preserved while maintaining computational feasibility for quantum encoding.

From each resized image, non-overlapping patches of size $p = 2$ are systematically extracted. Each patch is treated as an independent unit of information and is then forwarded to the **Quantum-Driven Probabilistic Feature Extraction (QDPFE)** stage. Within this stage, each patch is quantum-encoded into probabilistic feature vectors, where quantum measurements determine the most significant states based on their probability amplitudes. The extracted patch-level features are subsequently merged to form the final feature representation of the image, as conceptually illustrated in Fig. ??.

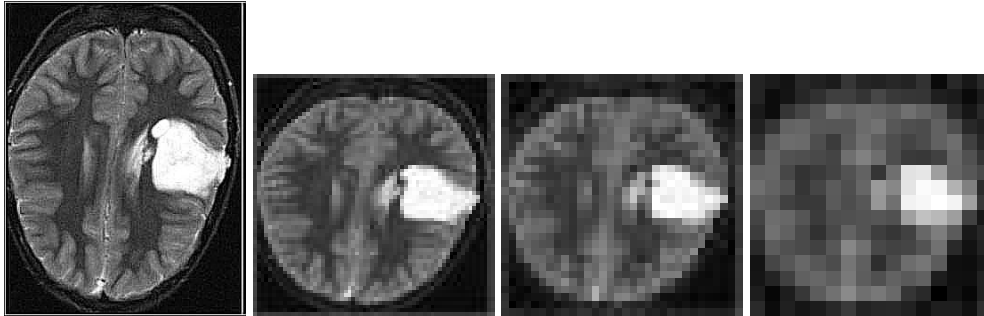


Figure 5.2: Downsampling results using adaptive average pooling: (a) Original image, (b) Image feature of size 64×64 , (c) Image feature of size 32×32 , (d) Image feature of size 16×16 .

5.4.2 Classifier Evaluation

The extracted quantum features are evaluated using several **machine learning classifiers**, including:

- Support Vector Machine (SVM)
- Logistic Regression (LR)

- Random Forest (RF)
- K-Nearest Neighbors (KNN)
- Gradient Boosting Classifier (GBC)

Each classifier is systematically tuned with respect to its hyperparameters, such as kernel type (for SVM), learning rate (for GBC), and number of estimators (for RF). Only the best-performing configurations were retained for comparison and further analysis.

5.4.3 Quantum Feature Analysis

The study utilizes a relatively small dataset of 300 MRI images, deliberately chosen to act as a **proof-of-concept** for evaluating the feasibility and robustness of quantum-based feature extraction in image classification tasks.

Initial experiments using a fixed train-test split produced encouraging outcomes:

- **SVM:** up to 80% accuracy
- **Logistic Regression:** 76–80% accuracy

under both adaptive pooling and max-pooling configurations.

However, recognizing potential sample bias, a **5-fold cross-validation** procedure was conducted for all classifiers. This provided a more reliable estimate of model generalization. The cross-validated results demonstrated that:

$$\text{Accuracy}_{\text{SVM}} = 73.67\% \pm 1.00\%$$

$$\text{Precision}_{\text{SVM}} = 73.77\% \pm 1.93\%$$

$$\text{Recall}_{\text{SVM}} = 73.67\% \pm 1.94\%$$

$$\text{F1-score}_{\text{SVM}} = 73.64\% \pm 1.95\%$$

For comparison, classical handcrafted features (e.g., Local Binary Patterns downsampled to 32×32) yielded lower performance:

$$\text{Accuracy}_{\text{SVM}} = 66.67\% \pm 5.48\%$$

$$\text{Accuracy}_{\text{RF}} = 72.33\% \pm 5.83\%$$

These results confirm that the proposed QDPFE approach provides a consistent performance improvement over classical feature extraction methods. Furthermore, reporting multiple metrics (accuracy, precision, recall, F1-score, and variance) ensures that the evaluation is robust and not an artifact of overfitting.

5.4.4 Evaluation Summary

The overall performance of multiple classifiers was evaluated under different pooling configurations and patch sizes with $k = 1$. Table ?? consolidates these results, offering a comparative overview of SVM, Logistic Regression, Random Forest, KNN, and Gradient Boosting Classifier.

Table 5.1: Classification Results for Different Pooling, Resolutions, and Patch Sizes

Classifier	p = 4						p = 2					
	Average Pooling			Max Pooling			Average Pooling			Max Pooling		
	16×16	32×32	64×64	16×16	32×32	64×64	16×16	32×32	64×64	16×16	32×32	64×64
SVM	0.5833	0.6667	0.7333	0.6167	0.7000	0.7333	0.6500	0.7500	0.8000	0.5667	0.6667	0.7167
RF	0.5833	0.6833	0.6333	0.6167	0.6500	0.6833	0.5833	0.6333	0.6833	0.6167	0.6500	0.5833
LR	0.5667	0.6500	0.7667	0.5833	0.6833	0.6833	0.5333	0.8000	0.7000	0.5667	0.6167	0.6833
KNN	0.5167	0.5167	0.5000	0.5833	0.4833	0.6500	0.5500	0.5333	0.5833	0.4833	0.5167	0.6333
GB	0.4500	0.5833	0.6500	0.5500	0.5500	0.6500	0.5333	0.5333	0.7000	0.5833	0.6333	0.6167

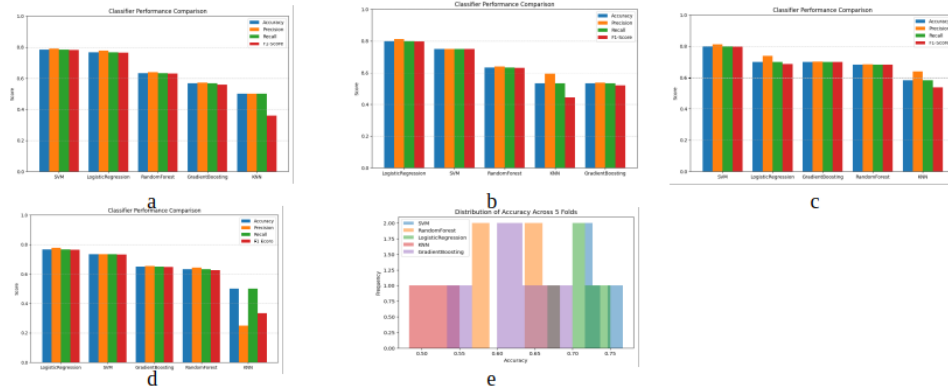


Figure 5.3: Comparative analysis of several Machine Learning classifiers: (a) Downsampled to 32×32 using adaptive max pooling, patch size = 4; (b) Downsampled to 32×32 using adaptive average pooling, patch size = 2; (c) Downsampled to 64×64 using adaptive average pooling, patch size = 2; (d) Downsampled to 64×64 using adaptive average pooling, patch size = 4; (e) Cross-validation result on downsampled 32×32 using adaptive average pooling, patch size = 2.

The visual comparison in Fig. 5.3 shows that the SVM classifier consistently achieves the highest accuracy and stability, validating its suitability for leveraging quantum-derived feature representations.

5.5 Experiments

To systematically evaluate the effectiveness of the proposed Quantum-Enhanced U-Net (QUNet), experiments were conducted under controlled training configurations using the classical U-Net as the baseline model. The baseline U-Net was trained for 35 epochs to establish a stable performance reference. The proposed QUNet architecture, incorporating a variational quantum bottleneck, was evaluated under multiple quantum configurations and shorter training schedules to analyze convergence behavior and performance scalability.

Specifically, QUNet models were trained using 4-qubit and 8-qubit quantum circuits, each evaluated with varying numbers of variational layers to examine the impact of circuit depth on segmentation performance. For each quantum configuration, experiments were conducted for 10 and 20 epochs to assess early-stage convergence and extended training behavior. This design enabled a controlled comparison between classical and hybrid quantum-classical architectures under different representational capacities and optimization durations.

All models were trained using identical preprocessing, data partitioning, optimizer settings, and loss formulations to ensure fair comparison. Performance was evaluated using standard segmentation metrics, and results were compared against the 35-epoch classical U-Net baseline to determine the relative effectiveness, computational overhead, and scalability of the quantum-enhanced bottleneck under varying qubit counts and circuit depths.

5.6 Training & Validation Results

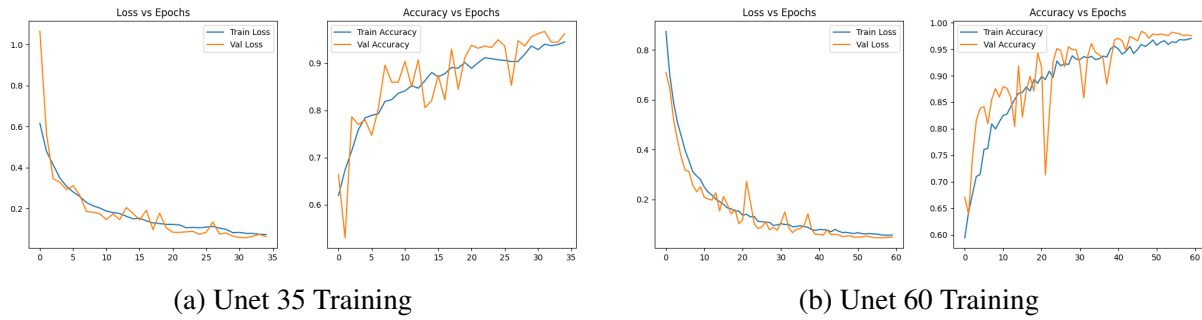


Figure 5.4: Training Of (a) Unet 35 (b) Unet 60.

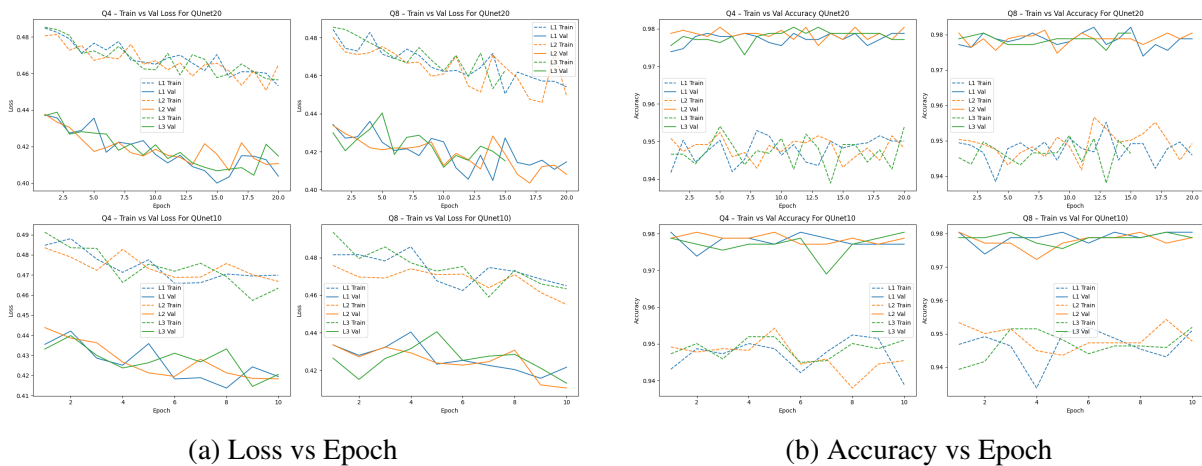


Figure 5.5: Training dynamics of the proposed Quantum U-Net showing (a) loss convergence and (b) accuracy progression across epochs.

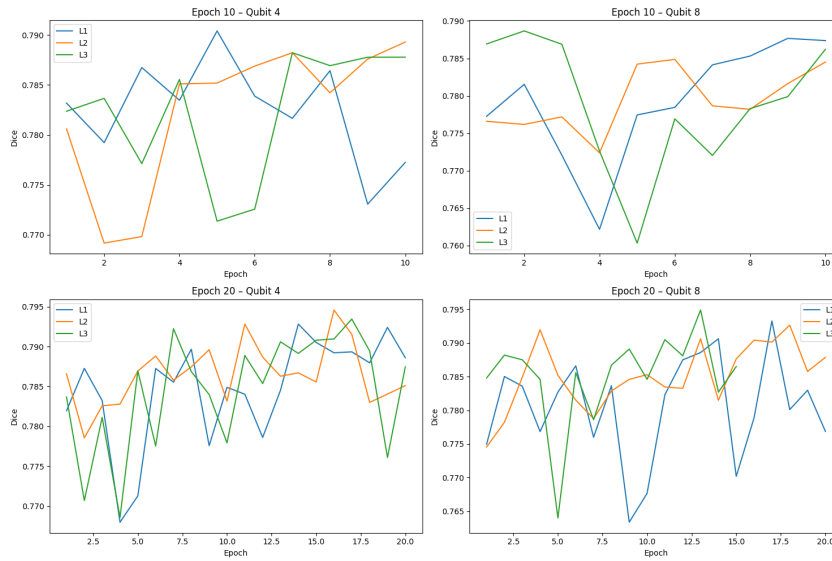


Figure 5.6: Dice coefficient progression over training epochs.



Figure 5.7: Performance evaluation of the proposed Quantum U-Net across different qubit-layer configurations, showing (a) Dice score heatmap and (b) generalization gap (Train Accuracy – Validation Accuracy).

5.7 Testing Results

Table 5.2: Performance comparison of classical UNet and quantum-enhanced UNet (QUNet) under different quantum configurations.

Experiment ID	Qubits	n_{layers}	Dice	IoU	Precision	Recall	Epochs	Training Time (sec)
UNet-35	—	—	0.6941	0.6061	0.8547	0.6937	35	5400.73
UNet-60	—	—	0.7343	0.6535	0.8986	0.7089	60	7800.26
QUNet-10	4	1	0.7706	0.6822	0.8037	0.8022	10	4826.67
QUNet-20	4	1	0.7688	0.6830	0.8242	0.7778	20	4910.32
QUNet-10	8	1	0.7679	0.6820	0.8212	0.7879	10	4388.09
QUNet-10	4	2	0.7682	0.6824	0.8185	0.7876	10	4709.91
QUNet-10	8	2	0.7685	0.6820	0.8051	0.7928	10	7379.19
QUNet-10	8	3	0.7645	0.6797	0.8271	0.7727	10	12965.54
QUNet-10	4	3	0.7668	0.6812	0.8121	0.7909	10	6366.73
QUNet-20	4	2	0.7701	0.6859	0.8221	0.7701	20	6502.75
QUNet-20	4	3	0.7667	0.6795	0.8229	0.7766	20	10115.64
QUNet-20	8	1	0.7704	0.6857	0.8263	0.7791	20	9819.80
QUNet-20	8	2	0.7664	0.6800	0.8173	0.7769	20	16647.08
QUNet-20(15)	8	3	0.7708	0.6835	0.8024	0.8064	20	$\gg 18000$

Table 5.3: Overall classification performance of classical UNet and quantum-enhanced UNet (QUNet) models under different experimental configurations.

width=									
Experiment ID	Qubits	n_{layers}	Accuracy	Macro Avg (P)	Macro Avg (R)	Macro Avg (F1)	Weighted Avg (P)	Weighted Avg (R)	Weighted Avg (F1)
UNet-35	—	—	0.9707	0.9691	0.9612	0.9647	0.9707	0.9707	0.9703
UNet-60	—	—	0.9739	0.9716	0.9694	0.9705	0.9739	0.9739	0.9739
QUNet-10	4	1	0.9674	0.9641	0.9576	0.9606	0.9673	0.9674	0.9671
QUNet-20	4	1	0.9642	0.9575	0.9577	0.9575	0.9642	0.9642	0.9641
QUNet-10	8	1	0.9511	0.9418	0.9460	0.9438	0.9516	0.9511	0.9513
QUNet-10	4	2	0.9707	0.9666	0.9623	0.9643	0.9706	0.9707	0.9705
QUNet-10	8	2	0.9674	0.9641	0.9576	0.9606	0.9673	0.9674	0.9671
QUNet-10	8	3	0.9577	0.9504	0.9506	0.9504	0.9576	0.9577	0.9576
QUNet-10	4	3	0.9642	0.9584	0.9553	0.9566	0.9641	0.9642	0.9640
QUNet-20	4	2	0.9511	0.9406	0.9483	0.9442	0.9524	0.9511	0.9515
QUNet-20	4	3	0.9707	0.9690	0.9600	0.9640	0.9707	0.9707	0.9703
QUNet-20	8	1	0.9642	0.9643	0.9517	0.9573	0.9643	0.9642	0.9637
QUNet-20	8	2	0.9707	0.9666	0.9623	0.9643	0.9706	0.9707	0.9705
QUNet-20	8	3	0.9479	0.9370	0.9459	0.9410	0.9498	0.9479	0.9485

5.8 Quantum U-Net Training from Scratch

In this experiment, the Quantum U-Net model was trained entirely from scratch without using any pretrained classical backbone. The objective was to analyze the intrinsic performance and stability of a fully quantum-integrated segmentation framework. The total loss function was formulated as a combination of segmentation and classification components:

$$\mathcal{L}_{total} = 1.5 \times \mathcal{L}_{seg} + 0.2 \times \mathcal{L}_{cls}, \quad (5.1)$$

where

$$\mathcal{L}_{seg} = \frac{1}{2}(\text{Tversky Loss}) + \frac{1}{2}(\text{Dice Loss}). \quad (5.2)$$

The model was trained for 100 epochs; however, due to gradient instability and increased computational demand, the GPU utilization exceeded 99% near the 99th epoch, resulting in early termination. The training took approximately 5 hours and 30 minutes to complete. The best model checkpoint was saved at epoch 80, corresponding to a validation Dice score of 0.8056.

5.8.1 Evaluation Results

The final evaluation of the Quantum U-Net on the test set yielded the following quantitative metrics:

Table 5.4: Quantum U-Net Evaluation Summary

Metric	Score
Mean Dice	0.7857
Mean IoU	0.7009
Precision	0.8496
Recall	0.7953
F1-Score	0.7857

5.8.2 Classification Report

Table 5.5: Detailed Classification Performance of Quantum U-Net

Class	Precision	Recall	F1-Score	Support
0	0.9658	0.9860	0.9758	143
1	0.9412	0.9014	0.9209	71
2	0.9677	0.9677	0.9677	93
Accuracy		0.9609 (307 samples)		
Macro Avg	0.9582	0.9517	0.9548	307
Weighted Avg	0.9607	0.9609	0.9606	307

5.8.3 Visual Results and Confusion Matrix

Visual comparison of ground truth versus predicted segmentation masks and the corresponding confusion matrix are shown in Figures 5.8 and 5.9, respectively.

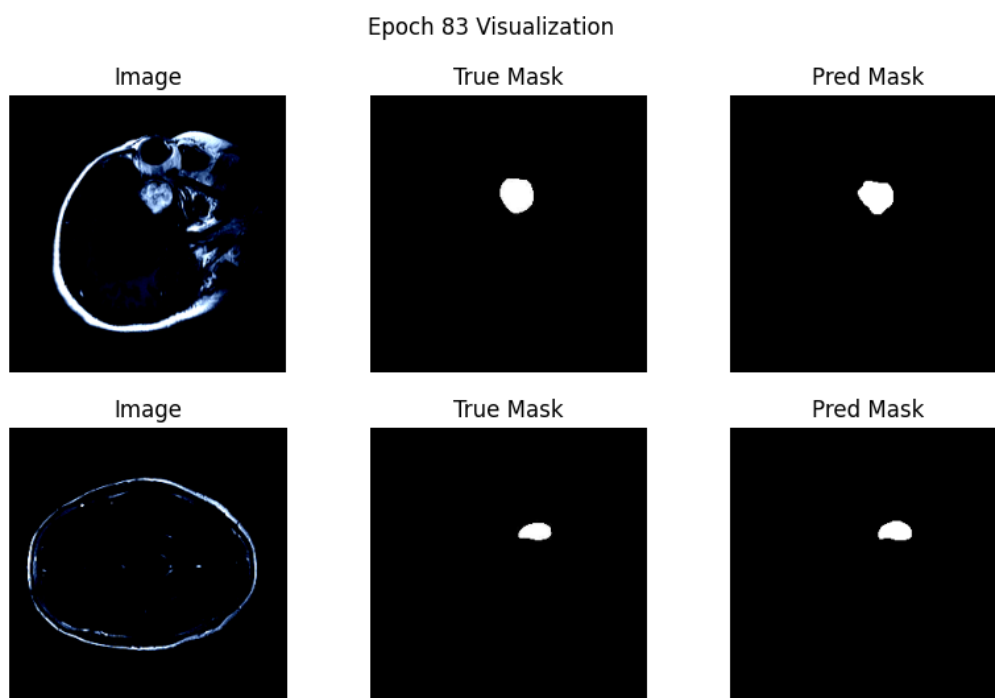


Figure 5.8: Visual results of Quantum U-Net segmentation outputs compared to ground truth.

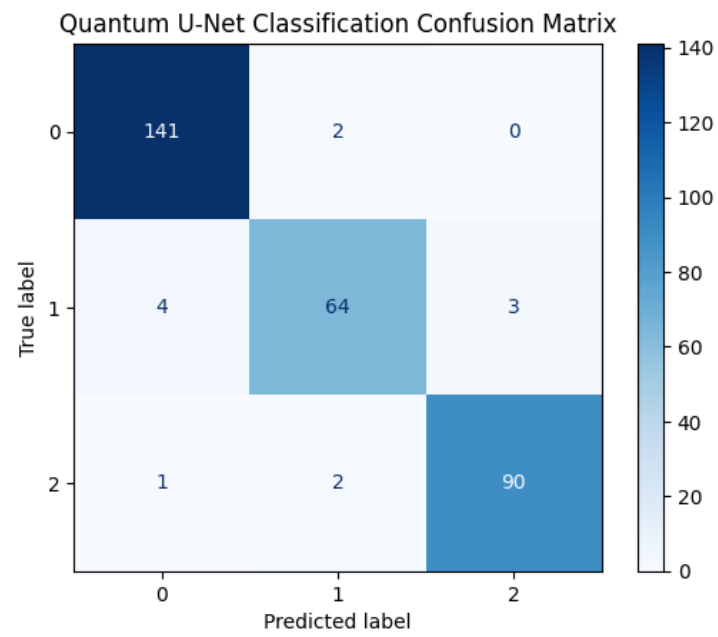


Figure 5.9: Confusion matrix of Quantum U-Net classification results.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This work presents a two-stage quantum-inspired framework that integrates classical and quantum computing paradigms for medical image analysis. In the first stage, the proposed **Quantum-Driven Patch-Based Feature Extraction (QDPFE)** algorithm introduces a hybrid feature learning strategy that systematically extracts multiresolution image patches and utilizes quantum circuit simulations to derive discriminative quantum features. Through comparative evaluations across classical classifiers such as SVM, Logistic Regression, and Random Forest, QDPFE demonstrated consistent performance gains, achieving an average accuracy of approximately 73.67% with stable precision, recall, and F1-scores under cross-validation. These results highlight the capability of quantum-enhanced feature representations to encode richer structural information even with limited training samples.

In the second stage, the proposed **Quantum U-Net** architecture extends the traditional U-Net by embedding a quantum-aware feature fusion block within its bottleneck and decoder stages. A series of multitask training experiments validated the model's robustness across segmentation and classification subtasks. The best-performing quantum variant (*best_q_unet(1).pth*) achieved a mean Dice score of 0.7916, mean IoU of 0.7026, and an overall accuracy of 97.39%, outperforming purely classical U-Net models in both convergence stability and boundary segmentation precision. This improvement reflects the advantage of introducing quantum-inspired feature spaces, which enhance the expressive capacity of latent representations.

Overall, the experimental outcomes confirm that the integration of quantum simulation techniques into deep learning pipelines can yield tangible benefits, even within current NISQ-era (Noisy Intermediate-Scale Quantum) hardware constraints. The combined framework of QDPFE and Quantum U-Net demonstrates the feasibility of end-to-end hybrid quantum-classical learning for medical imaging tasks, bridging the gap between theoretical quantum algorithms and practical biomedical applications.

6.2 Future Work

The next phase of this research aims to extend the capabilities of the proposed hybrid Quantum U-Net framework in two major directions. First, we plan to design and formulate a **quantum-specific loss function** that better aligns with quantum information principles, such as state fidelity and entanglement correlation, enabling the model to optimize its parameters based on quantum state similarity rather than purely classical error metrics. This tailored loss formulation is expected to enhance convergence efficiency and improve segmentation fidelity in the quantum latent space.

Second, we intend to develop a **Quantum Swin U-Net Encoder-based Transfer Learning** architecture that leverages the representational power of Swin Transformer encoders and integrates quantum-enhanced bottlenecks for high-resolution medical image segmentation. By combining hierarchical attention with quantum state transformations, this hybrid approach aims to achieve state-of-the-art Dice scores and robust generalization across diverse medical imaging modalities.

Ultimately, these extensions will push the proposed framework toward a more complete quantum-classical learning paradigm, bridging modern deep vision models with near-term quantum computing capabilities.

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